Catena 234 (2024) 107613

Contents lists available at ScienceDirect

Catena

journal homepage: www.elsevier.com/locate/catena

Soil water infiltration characteristics of reforested areas in the paleo-periglacial eastern Liaoning mountainous regions, China

Di Wang [a], Jianzhi Niu [a,*], Tao Yang [a], Yubo Miao [a], Linus Zhang [b], Xiongwen Chen [c], Zhiping Fan [d], Zhengyu Dai [a], Haoyang Wu [a], Shujian Yang [a], Qihuang Qiu [a], Ronny Berndtsson [b]

[a] School of Soil and Water Conservation, Beijing Forestry University, No. 35 Tsinghua East Road, Haidian District, Beijing 100083, China

[b] Division of Water Resources Engineering & Centre for Advanced Middle Eastern Studies, Lund University, Box 118, Lund SE-221 00, Sweden

[c] Department of Biological and Environmental Sciences, Alabama A&M University, Normal, AL 35762, USA

[d] School of Environmental & Safety Engineering, Liaoning Petrochemical University, Fushun 113001, China

ARTICLE INFO

Keywords:
Infiltration capacity
Reforestation approaches
Soil properties
Root abundance (*RA*) and gravel distribution
Soil water content (*SWC*)
Paleo-periglacial landform

ABSTRACT

Plantation forests (PF) and natural secondary forests (NSF) are the primary reforestation approaches. The establishment of PF can affect forest hydrological processes by changing soil structure. To date, few studies have focused on these changes and the effects on hydrological processes for the paleo-periglacial landform. To reveal reforestation approaches effects on water infiltration, including soil water infiltration capacity, retention capacity, and waterflow path pattern, we conducted field dye-tracer investigations with rainfall and laboratory infiltration experiments for the paleo-periglacial landform of eastern Liaoning mountains, China. The results showed that (1) Soil physical properties (including total porosity (*TP*), capillary porosity (*CP*), non-capillary porosity (*NCP*), initial soil water content (*IWC*), field water capacity (*FWC*)) and root abundance (*RA*) decreased with soil depth in both PF and NSF, while the soil bulk density (*BD*) and distribution of gravel content showed opposite changes. (2) Establishment of PF reduced the infiltration capacity and water retention capacity in the 0–20 cm layer, but enhanced the water retention capacity in 20–30 cm layer. Low *IWC* was conducive to increase soil water content (*SWC)* after infiltration. (3) Infiltration capacity parameters (including saturated hydraulic conductivity (*Ks*), *SWC*, difference between *SWC* and *IWC* (*SWC–IWC*), dye coverage ratio (*DC*)) were significantly correlated with *BD*, *TP*, *CP*, *NCP*, *FWC*, and fine roots *RA* ($P < 0.05$). Better connectivity gravels were more conducive to water infiltration. (4) Preferential flow was the main infiltration type, but with different waterflow paths pattern, with the 'funnel', 'finger' shape for PF, NSF, respectively. Increasing infiltration could increase flow path connectivity. Our findings show that soil physical properties, roots, and gravel occurrence affected soil infiltration, and different reforestation approaches had varying impacts on soil infiltration, water redistribution, transportation, and storage of surface and groundwater, improving the understanding of ecohydrological processes and effects of water resources management in forest ecosystems of paleo-periglacial landform.

1. Introduction

Soil infiltration is an important component in the process of the water cycle and runoff formation (Duan et al., 2021), and a key factor restricting runoff generation and vegetation growth (Alaoui et al., 2011). Soil water infiltration characteristics and soil water status reflect the ability of soil to supply water to vegetation, groundwater recharge, and erosion susceptibility to rainwater and runoff (Brid et al., 1996). Therefore, soil infiltration is essential for the water balance, soil erosion, water pollution, geohazard prediction, and management of forest ecosystems (Huang et al., 2017; Lipiec et al., 2006).

Saturated hydraulic conductivity (K_S), a vital hydraulic parameter in the study of soil water infiltration and solute transport as it reflects the saturated permeability and saturated water flux of soil in natural field state and affects the storage and transport of soil water (Gwenzi et al., 2011). High K_S can accelerate rainfall infiltration, thereby reducing surface runoff and soil erosion (Chen et al., 2021). In addition, soil water content (*SWC*) is also a critical parameter in the characterization and

* Corresponding author at: School of Soil and Water Conservation, Beijing Forestry University, No. 35 Tsinghua East Road, Haidian District, Beijing 100083, China.
E-mail address: nexk@bjfu.edu.cn (J. Niu).

https://doi.org/10.1016/j.catena.2023.107613
Received 17 July 2023; Received in revised form 16 September 2023; Accepted 19 October 2023
Available online 31 October 2023

modeling on soil water flux (Dubus and Brown, 2002). Initial soil water content (*IWC*) is a key factor influencing soil water transport and solute transport, but due to the stochastic and complex nature of water flow, the relationship between them remains inconclusive (Merdun et al., 2008). Vegetation type (Qiu et al., 2023), physical properties of soil (including bulk density (*BD*), total porosity (*TP*), capillary porosity (*CP*), non-capillary porosity (*NCP*), *IWC*, etc.) (Gumbs and Warkentin, 1972; Kan et al., 2020; Qiu et al., 2023) and the soil macropore channels formed by vegetation roots (Jiang et al., 2018), soil animals (Murphy and Banfield, 1978), gravels (Meyer et al., 1972), etc., are all critical elements on soil infiltration. In addition, global climate change and extreme weather such as rain bursts have an important impact on forest hydrological processes (Luo et al., 2022). In general, most of the existing studies on factors influencing soil water infiltration have focused on single or several factors such as soil properties, vegetation types, root distribution characteristics, gravel characteristics and rainfall, while there are few studies that combine all these variables at different scales.

Due to the differences in surface root-gravel structure of different types of forest land, the distribution of soil macropores is different, which in turn has a crucial influence on soil water infiltration and retention (Meyer et al., 1972; Porporato et al., 2002; Rodríguez et al., 2018). Plant root system, a key element in the vegetation-soil-hydrologic ecosystem, provide water channels for soil infiltration to move into deeper soil (Cui et al., 2019). Furthermore, studies have shown that not all roots promote vertical movement of water. Some roots may even impede vertical infiltration and facilitate lateral water flow (Luo et al., 2019). Root distribution (Luo et al., 2019), diameter class (Ghestem et al., 2011; Guo et al., 2019), number (Jørgensen et al., 2002), and other root architectural traits (i.e., curvature, length, extension direction, etc.) (Ghestem et al., 2011; Pierret et al., 2007; Stokes et al., 2009) affect the water infiltration process (Schwärzel et al., 2012; Zhang et al., 2022). The difference in root architectural characteristics of different vegetation types can lead to different patterns and effects on water infiltration (Jiang et al., 2018). In addition, the size and position of gravel (Kato et al., 2009; Nyssen et al., 2006) play large role in the transport and redistribution of soil water (Nasri et al., 2015). At present, there is no consensus on how the distribution of gravel affects soil water and solute transport process (Zhou et al., 2011). Previous studies have indicated that K_S decreases with the increase of soil gravel content for the range 0 ~ 40 % (Novák et al., 2011; Zhou et al., 2009). However, it has also been shown that K_S tends to increase and then decrease with increasing gravel content (Ma and Shao, 2008). Regarding gravel size, Ma and Shao (2009) suggested that 2–3 mm gravel could impede water infiltration, while gravel larger than 25 mm have a facilitating effect for soil water transport. However, Guo et al. (2010) concluded the opposite effect, that gravel with small particle size can play a more positive role in water infiltration. In-situ observations of the influence of root and gravel distribution on hydrological processes is a key issue in studies of hydrological processes in forest ecosystems. At present, there is still a lack of research on the effects of root and gravel spatial distribution on water infiltration at a larger scale.

Periglacial landform refers to the land type developed by freeze–thaw action in harsh climate areas not covered by glaciers, which was mainly produced in the peak period of Quaternary Glaciation. Approximately 20 % of the world's continental area are covered by periglacial environments (Gutiérrez and Gutiérrez, 2016). China's periglacial landforms are mainly distributed in the Changbai Mountains, the Northeast Greater and Lesser Khingan Mountains, the western alpine plateau, and some higher mountainous areas in the east (Cui and Zhu, 1988). The Laotudingzi Mountain area in eastern Liaoning Province, a remaining cluster of the Changbai Mountain, has formed a variety of paleo-periglacial landform types due to periglacial forces of the Quaternary Glaciation. The site is prone to major geological hazards such as mountain tsunamis and debris flow with fragile habitats. The gravel is disorderly overlapped with infertile soil and poor substrate stability, unstable rooting of vegetation, with frequent heavy rainstorms in the rainy season and susceptible to soil and water erosion (Liu et al., 2014; Zhang et al., 2015). In addition to natural secondary forests, large areas of plantation forests have been established in this region, to improve water conservation, providing high-quality water for downstream cities, preventing geological disasters, and ecological quality of the watershed in general. Plantation forests can regulate rainfall redistribution and improve *SWC* (Demir et al., 2022). In the study area, plantation forests have better vegetation stability even with a single type of tree species compared with natural secondary forests. Currently, natural secondary forests and plantation forests are common approaches of reforestation (Hua et al., 2022). Reforestation can positively affect ecosystem function and maintain ecosystem services (Barral et al., 2015), it also has a significant impact on soil infiltration. It has been shown that reforestation can enhance the infiltration capacity of degraded soils (Lozano-Baez et al., 2019; Filoso et al., 2017). It is generally recognized that reforestation significantly alters the physiochemical properties of the soil and hence the infiltration capacity (Deng et al., 2013; Qiu et al., 2023). Therefore, it is of great significance to understand the hydrological processes of surface soil for different reforestation types. However, this is still poorly understood. Especially in the paleo-periglacial landform area, which is a major obstacle for sustainable development, biodiversity, and potential management of forest watersheds.

Considering the essential role of reforestation processes and the lack of research on soil infiltration characteristics of forest land in paleo-periglacial landform areas, the objective of this study was to assess the impact of reforestation on soil water infiltration and water retention capacity in typical paleo-periglacial landforms in China. We hypothesize that (1) the establishment of plantation forests in reforestation approaches changes the infiltration capacity and enhances the water retention capacity of the soil; (2) Hydrological processes in different reforestation approaches are closely related to different soil characteristics, roots, and gravel distribution; (3) Increasing infiltration will enhance soil infiltration capacity and hydrological connectivity. The findings will help to understand the forest hydrological processes of periglacial geomorphology in humid area, and be of guidance for water resources management, watershed management, and ecological reforestation in the paleo-periglacial landform of eastern Liaoning mountains and other periglacial landscapes.

2. Materials and methods

2.1. Site description

The study area, the Laotudingzi National Nature Reserve (41°11′11″–41°21′34″N, 124°41′13″–125°05′15″E) in northern Liaoning Province, China, is located at the junction of Xinbin Manchu Autonomous County and Huanren Manchu Autonomous County (Fig. 1). The study area belongs to the remnants of the Changbai Mountains, which is a typical paleo-periglacial geomorphic area. It belongs to the cold and humid climate zone of eastern Liaoning in the north temperate continental monsoon climate. And according to the World Map of Köppen-Geiger climate classification (Kottek et al., 2006), the climate class of the region belongs Dwa (Snow climate with dry winter and hot summer). The annual average temperature is 6.0 °C, absolute minimum temperature is –33.2 °C, absolute maximum temperature is 38 °C, and the average frost-free period is 139 days. Rainfall is abundant, mainly from June to August, with an average annual precipitation of 827.8 mm and an average relative humidity of 73 %, making it one of the wettest areas in Liaoning Province. The soil types are brown soil and dark brown soil, while the main soil texture class according to WRB (2015) is sandy loam. The basic soil physical properties are sand 58 %–72 %, silt 15 %–29 % and clay 8 %–12 %, the basic chemical properties show that pH 5.5–6.2, soil organic carbon content is about 76.7 $g \cdot kg^{-1}$ –156.3 $g \cdot kg^{-1}$. Due to the specificity of the landform type, disordered overlapping gravel soil is widely distributed, mostly developed from granite residual parent material. The thickness of the soil layer is unevenly

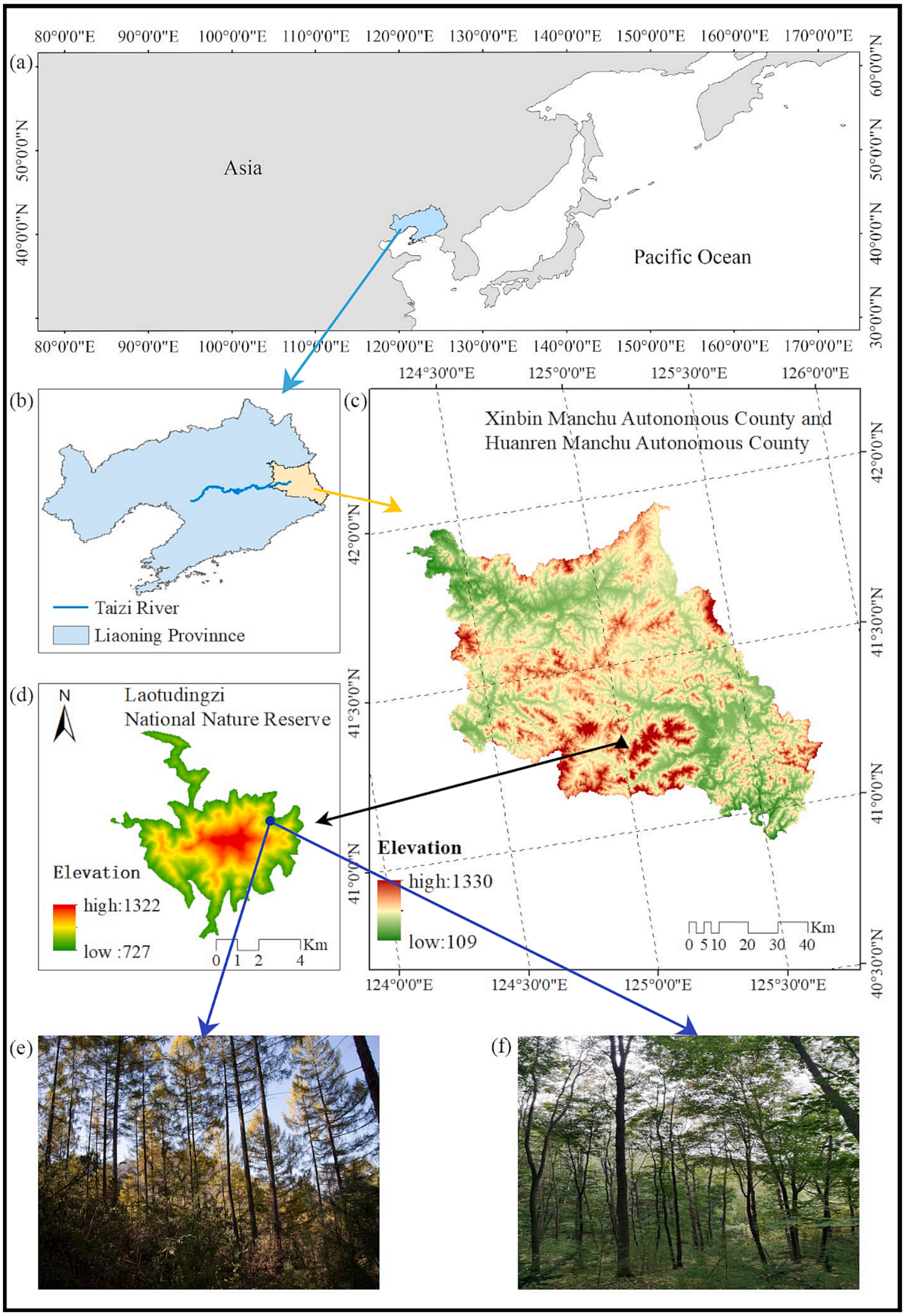

Fig. 1. Geographical location of the study area. (a), (b), (c) show the geographical location of the study area in different map ranges. (d) shows the Laotudingzi National Nature Reserve. The two approaches of reforestation include (e) plantation forests (PF) and (f) natural secondary forests (NSF).

distributed, usually 5–50 cm. The soil matrix has poor stability, wet soil, high organic matter content, and sufficient nutrients for the growth and development of forest plants. The canopy density of natural secondary forests and plantation forests in the study area is very high, mostly above 0.8, but the vegetation is unstable and prone to soil erosion. In addition, this area is the headstream of the Taizi River, which is an important source of water for the urban and rural population living in the lower reaches of Liaoning Province. The forest land in this area is planned to be a water conservation forest, and the construction of water conservation forest has become the main task of the forest ecosystem construction in the region.

In this research we selected two typical types of reforestation approaches sites, natural secondary forest (NSF) and plantation forest (PF), which are widely distributed in the study area. To reduce the experimental error, altitude, slope, and aspect of the two forest plots we selected were essentially the same. The specific sites information is shown in Table 1. The experiments were carried out in October 2021.

Table 1
Basic information of investigated sites.

Plot	reforestation approach	Major tree species	Age /year	Longitude /°	Latitude /°	Average DBH/cm	Average Tree Height /m	Coverage /%	Slope /°	Vegetation composition
PF	plantation forest	*Larix gmelinii, Pinus koraiensis*	47	124.816	41.304	26.46	22.31	75	16–19	Associated tree species are *Phellodendron amurense* and *Juglans mandshurica*, with a very small proportion. Vines is *Actinidia arguta*. Groundcovers are *Leymus chinensis, Geranium wilfordii, Glycine soja*, etc.
NSF	natural secondary forest	*Quercus mongolica, Acer pictum, Populus davidiana*	47	124.815	41.305	26.32	21.46	92	15–18	Associated tree species are *Kalopanax septemlobus, Carpinus cordata*. Groundcovers are *Rubia chinensis, Urtica angustifolia, Viola phillipina, Viola arcuata*, etc.

2.2. Soil physical properties

Scattered soil samples were taken in 10 cm layers in the vicinity of each of the NSF and PF dye-tracer experiment sites to measure the physical properties of the soil. Soil samples were collected at three depths of 0–10 cm, 10–20 cm, and 20–30 cm using cutting cylinders (volume, 100 cm^3) at each site, with three replicates per layer, and weighing mass of the cutting cylinder with initial wet soil (M_{wet}, g). And placed the cutting cylinders in distilled water for 24 h to saturate, weighed with saturated soil water (M_{24h}, g). The cylinders were placed on coarse sand to drain by gravity and then weighed after 2 h (M_{2h}, g) and 4 days (M_{4d}, g). Next, the samples were dried in an oven at 105° C for 24 h and weighed (M_{dry}, g). Finally, the empty cutting cylinder was weighed (M_{cemp}, g). Calculation of soil porosity was done using undisturbed soil columns after saturation, assuming that the soil pores are filled with water without air and verified by dry bulk weight and particle density of 2.65 g·cm^{-3} (Danielson & Sutherland, 1986). *BD, TP, CP, NCP*, *IWC*, and field water capacity (*FWC*) were calculated according to:

$$BD\left(\mathrm{g{\cdot}cm^{-3}}\right) = \frac{M_{dry}(g) - M_{cemp}(g)}{100(\mathrm{cm^3})} \tag{1}$$

$$TP\,(\%) = \left(1 - \frac{BD(\mathrm{g{\cdot}cm^{-3}})}{2.65}\right) \times 100\% \tag{2}$$

$$CP\,(\%) = \frac{M_{2h}(g) - M_{dry}(g)}{100 \times \rho_{water}(\mathrm{g{\cdot}cm^{-3}})} \times 100\% \tag{3}$$

$$NCP\,(\%) = TP(\%) - CP(\%) \tag{4}$$

$$IWC\,(\%) = \frac{M_{wet}(g) - M_{dry}(g)}{M_{dry}(g) - M_{cemp}(g)} \times 100\% \tag{5}$$

$$FWC\,(\%) = \frac{M_{4d}(g) - M_{dry}(g)}{M_{dry}(g) - M_{cemp}(g)} \times 100\% \tag{6}$$

2.3. Dye-tracer experiment

Dye-tracer experiments were carried out at the PF and NSF sites, each with a relatively flat surface of 1.1 m × 1.2 m and no significant waterlogging, to determine plot-scale water infiltration patterns and flow paths. To prevent the influence of main trunk of trees on infiltration, the sites were selected at approximately equal distances from the surrounding trees. Before dying, we carefully cleaned the areas from surface litter and weeds. Subsequently, a stainless steel metal frame was pushed into the soil at a depth of approximately 10 cm with a rubber hammer, and the soil around the metal frame was compacted to avoid edge effects.

We collected rainfall data for two sites around the study area for 2015–2020 from the China Meteorological Data Service Centre (https://data.cma.cn/site/index.html). The rainfall data were divided into four types: light rainfall (LR, daily precipitation in the range of 0–10 mm), moderate rainfall (MR, daily precipitation in the range of 10–25 mm), heavy rainfall (HR, daily precipitation in the range of 25–50 mm), and extreme rainfall (ER, daily precipitation over 50 mm) (Ma et al., 2023) (Fig. 2). The total precipitation for HR and ER accounted for 31 % and 16 % of all rainfall, respectively. Surface runoff and soil erosion generally occur under intensive rainfall conditions, thus, we selected HR and ER as the experimental infiltration intensity in this study. Combined with the average rainfall of each rainfall intensity, we set the HR and ER infiltration amount for dye-tracer experiment to 35 mm and 60 mm, respectively.

To avoid effects of rainfall, the dye infiltration study was conducted during a dry period with no rainfall 3 days before and 3 days after the experiments with a layout according to Fig. 3. We used 5 g·L^{-1} Brilliant Blue FCF dye solution (Luo et al., 2022; Zhu et al., 2019). A backpack electric sprayer was used to apply the dye solution at a rate of 40 mm·h^{-1} in each field with a spray uniformity coefficient of 0.95 to ensure no significant water accumulation throughout the infiltration process. After the dye-tracer experiment, the infiltration sites were covered with waterproof cloth to prevent surface evaporation.

The waterproof cloth and metal frame were carefully removed 24 h after the dye-tracer experiment. Vertical soil profiles were dug according to Fig. 3, with 5 cm interval. Two steel rulers were used as markers, placed at two adjacent sides of each profile in both horizontal and

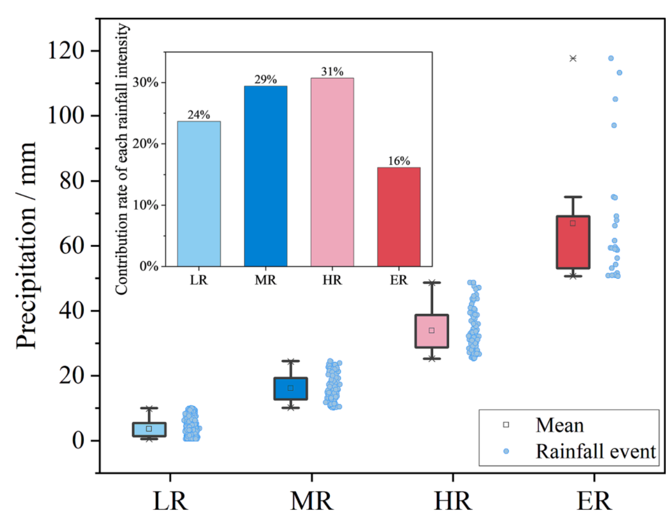

Fig. 2. Rainfall intensity and distribution in the study area from 2015 to 2020. LR, light rainfall; MR, moderate rainfall; HR, heavy rainfall; ER, extreme rainfall.

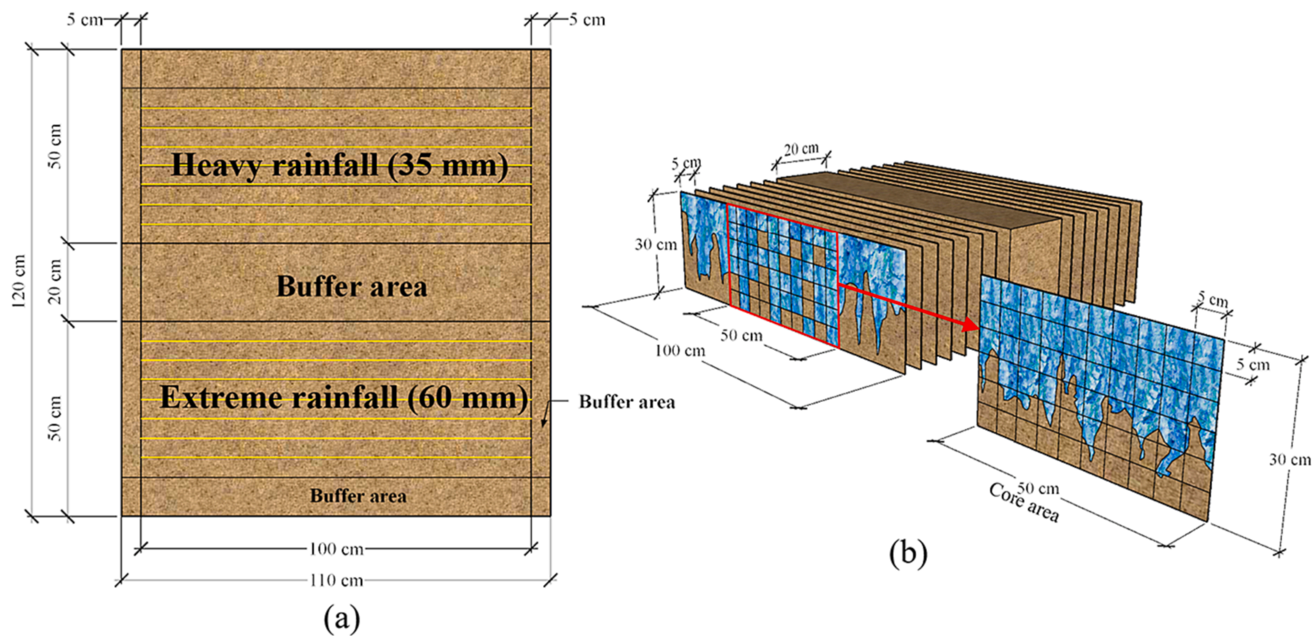

Fig. 3. Dye-tracer experimental set-up and soil sample collection layout. (a) top view of the dye-tracer sites, with the yellow horizontal line representing the location of the sampled vertical soil profiles. (b) vertical profile sampling layout, with the core area (30 cm × 50 cm) in red frame, where each grid represents the smallest unit sample (5 cm × 5 cm). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

vertical direction, and a digital camera (Cannon 700D) was used to photograph the infiltration pattern of each 5 cm interval profile. Each site had 7 profiles, and at least 3 photos were taken for each profile. According to Fig. 3 (b), 5 cm × 5 cm was taken as the smallest unit to sample in the core area of each profile. Accordingly, 60 samples were taken from each profile, and each sample was about 70 g. Thus, a total of 1680 soil samples were taken to measure the *SWC*. The profile soil samples were weighed (M_{dye_wet}, g), and then dried in an oven at 105 °C for 24 h and weighed (M_{dye_dry}, g), and *SWC* was calculated according to:

$$SWC\,(\%) = \frac{M_{dye\_wet}(g) - M_{dye\_dry}(g)}{M_{dye\_dry}(g)} \times 100\% \tag{7}$$

The dye images were spatially corrected according to scale in Photoshop 2020 using the "perspective cutting tool" and the images of the profile area to be analyzed were cropped. We used the "adjustment - brightness/contrast", "color replacement tool" to replace the dyed areas with black and the undyed areas with white. In addition, gravel protruding from the profile was selected and replaced with white, the images were exported to grayscale images with a resolution of 1 pixel·mm^{-1}. In addition, due to the high gravel content of the paleo-periglacial landform in the study area, all gravels (including gravels protruding from the profile, and considered these gravels as undyed gravels) in the profile were divided according to dyed and undyed gravels, and gravel images were derived according to the image processing methods described above. The number of pixel points occupied by different areas was finally obtained by MATLAB 2020, and the area of area (D, mm^2) and the undyed area (ND, mm^2) in the profile and gravel images were quantified and analyzed separately. Furthermore, a gravel three-dimensional spatial distribution was constructed from the gravel area data by the spatial difference method. The profile dye coverage ratio (*DC*, %) was calculated according to (Flury et al., 1994):

$$DC\,(\%) = \left(\frac{D(mm^2)}{D(mm^2) + ND(mm^2)}\right) \times 100\% \tag{8}$$

2.4. Root abundance (RA) investigation

In the minimum unit grid in the core area of each soil profile, the root abundance (*RA*) was investigated by dividing the root points into coarse roots (d ≥ 5 mm) and fine roots (d < 5 mm) according to the root point diameter (Van Noordwijk et al., 2000). The number of root points at different diameter levels was counted according to whether they were in the dyed area or not.

2.5. Infiltration measurements

Considering the small portion of undyed soil area in each layer for the 60 mm infiltration, the K_S characteristic difference between dyed and undyed area was determined for the 35 mm infiltration site (PF$_{35}$, NSF$_{35}$). To reduce experimental error considering the size of the cutting cylinder and the spacing of the profiles, we sampled the 1st, 3rd, and 5th profiles with a volume of 100 cm^3 cutting cylinder at 10 cm intervals with soil depth. In each profile, one soil sample was taken from each dyed and undyed area at each soil depth, for a total of 54 samples. The obtained cutting cylinder soil sample was placed in distilled water for 12 h to make it saturated, and the K_S was measured by the constant head method (Wang et al., 2023). The height of the water head was controlled to be 2 cm using a Markov bottle. The outflow was recorded each 5 s during the first 1 min, and then each 10 s until the outflow rate was constant. *Ks* was calculated using Darcy's law:

$$K_s\left(mm{\cdot}min^{-1}\right) = \left(\frac{Q(ml) \times L(mm)}{S(mm^2) \times t(\min) \times h(mm)}\right) \times 100\% \tag{9}$$

where K_s, saturated hydraulic conductivity; Q, flow through the unit cross-section; L, soil sample thickness; S, the cross-sectional area of the cutting cylinder; t, infiltration time; h, constant head height.

Since soil K_S is affected by the difference in viscosity at different water temperature, calculated K_S was converted to K_S at 10 °C (K_{10}) by:

$$K_{10}\left(mm{\cdot}\min^{-1}\right) = \left(\frac{K_s(mm{\cdot}\min^{-1})}{0.7 + 0.03T(^\circ C)}\right) \times 100\% \tag{10}$$

where K_{10}, K_S at 10 °C; T, water temperature at the time of measurement. Note that the K_S analyzed in this study refers to the K_S at 10 °C.

2.6. Other relevant measurements

Like the sampling principle in 2.3, to control errors and uncertainties and facilitate correlation analysis, random samples were taken at 10 cm intervals in the 1st, 3rd, and 5th profiles of the 35 mm infiltration sites (PF_{35}, NSF_{35}) with a 100 cm^3 cutting cylinders, 3 samples per profile, in total 18 soil samples. Subsequent soil physical properties (Eq. (1)–(6)) were determined and correlation analysis of soil properties with water infiltration parameters was performed.

2.7. Data analysis

One-way ANOVA was used to evaluate the effects in different soil layers and reforestation methods from soil physical properties. The effects at different soil depths from infiltration capacity (K_S) in dyed and undyed areas were assessed with one-way ANOVA. The least significant difference (LSD) method was used to test the differences in soil properties between different reforestation approaches and different soil layers. Shapiro-Wilk test was used to test the normality of data, and log-transform (log base 10) was used to convert non-normally distributed data into normally distributed data. Differential analysis of gravel content, *RA*, *DC*, and *SWC* under different reforestation approaches and different infiltration conditions, if the data conformed to or transformed into normal distribution, independent-sample *t* test ($P < 0.05$) was used for evaluation. Otherwise, we used nonparametric Mann-Whitney *U* test ($P < 0.05$). The relationship between soil properties and infiltration parameters was determined by Pearson correlation analysis and linear regression analysis. All statistical analyses were performed using IBM SPSS Statistics 22.0 and all figures were created by Origin 2023 software.

3. Results

3.1. Soil characterization

3.1.1. Soil physical properties

Table 2 shows the main soil physical characteristics of soil at different depths and reforestation type. *BD*, *TP*, *CP*, *IWC*, and *FWC* were significantly affected by soil depth ($P < 0.05$), and *BD* and *NCP* were significantly affected by the type of forest land ($P < 0.01$). Furthermore, the effects of the interaction between depth and forest land type on *BD*, *TP*, *CP*, *NCP*, *IWC*, and *FWC* were statistically significant ($P < 0.01$). The *BD* in PF was significantly higher than that in NSF at different soil depths ($P < 0.05$), which means that artificial interference substantially increased *BD*. For soil depth, *BD* increased with increasing depth. The highest value of *BD* (1.58 $g \cdot cm^{-3}$) was found in PF at the depth of 20–30 cm. Correspondingly, the *TP* and *NCP* of NSF was higher than that of PF ($P < 0.05$), especially at 0–20 cm soil layer. *TP*, *CP*, and *NCP* gradually decreased with soil depth. Note that the *TP* and *CP* in NSF changed more than those in PF. The *IWC* and *FWC* of NSF were higher than that of PF, in addition to the 20–30 cm layer *IWC*. No significant differences were found between the two sites for *FWC*. Both FWC and *IWC* decreased in decreasing order as soil depth, and the surface layer (0–10 cm) *IWC* was significantly higher than 10–30 cm ($P < 0.01$), the *FWC* at 20–30 cm was significantly lower than that at 0–10 cm ($P < 0.01$). The *IWC* and *FWC* were higher by 34.1 % and 60.8 % ($P < 0.01$) in NSF at 0–10 cm, respectively, than at 20–30 cm. The *IWC* and *FWC* at 0–10 cm were, respectively, higher by 17.2 % and 55.9 % than at 20–30 cm in PF ($P < 0.01$).

3.1.2. Gravel distribution

The soil was characterized by extensive gravel contents. Gravel may play an important role for soil infiltration (Valentin and Casenave, 1992; Poesen, 1986). Thus, we analyzed the three-dimensional spatial distribution characteristics of gravel in each plot based on the dyed profile images of the soil (Fig. 4(a)-(d)). Correspondingly, the variation of gravel content with depth is shown in Fig. 4(e)-(h). As can be seen from Fig. 4, the distribution of gravel at each site was spatially heterogeneous, with maximum contents ranging 22.8 % (NSF_{60})–53.1 % (PF_{35}). The gravel content showed a tendency to increase with soil depth, and there were significant differences in gravel content between soil layers ($P < 0.01$). The gravel content in PF was significantly higher than that in NSF ($P < 0.01$), and the gravel in PF was larger than that in NSF, while the fine gravel in NSF was relatively more (Fig. 4).

3.1.3. Root abundance (RA)

The distribution of *RA* with soil depth for total, fine (d < 5 mm), and coarse roots (d ≥ 5 mm) in the PF and NSF sites are shown in Fig. 5. In the 0–30 cm soil layer, the coarse roots were few, accounting for only 1.6 % (PF) and 4.0 % (NSF) of total root *RA*. Overall, the distribution of *RA* for total and fine roots was basically identical, as evidenced by the decrease in *RA* with soil depth. The distribution of *RA* in total and fine roots was significantly different among different soil layers ($P < 0.01$). Note that the total and fine roots in each layer of the NSF site were significantly higher than those of the PF ($P < 0.01$). The *RA* of total and fine roots in the PF site decreased by 92.7 % and 92.9 % from surface to

Table 2
Characteristics of soils at different depth and forest land type.

Depth (cm)	Type of forest land	*BD* ($g \cdot cm^{-3}$)	*TP* (%)	*CP* (%)	*NCP* (%)	*IWC* (%)	*FWC* (%)
0–10	PF	**1.37 (0.02)** Aa	56.15 (2.37) Ba	36.25 (1.30) Ba	19.90 (1.07) Ba	27.04 (0.96) Ba	48.06 (4.19) Aa
	NSF	0.94 (0.12) Bb	**64.59 (1.88)** Aa	**41.24 (2.76)** Aa	**23.34 (1.53)** Aa	**29.60 (1.09)** Aa	**53.60 (2.29)** Aa
10–20	PF	**1.46 (0.07)** Aa	47.41 (2.25) Bb	**30.75 (1.36)** Ab	16.66 (1.03) Bb	24.15 (1.26) Ab	43.48 (2.98) Aab
	NSF	1.25 (0.07) Ba	**52.88 (2.06)** Ab	30.60 (2.64) Ab	**22.28 (0.69)** Aa	**24.26 (0.86)** Ab	**49.53 (6.59)** Aa
20–30	PF	**1.58 (0.19)** Aa	42.53 (1.97) Ab	**26.57 (2.52)** Ab	15.97 (1.10) Ab	**23.07 (0.75)** Ab	30.83 (3.49) Ab
	NSF	1.39 (0.16) Aa	**45.05 (1.74)** Ac	25.03 (0.64) Ab	**20.02 (2.37)** Aa	22.07 (0.88) Ab	**33.33 (5.31)** Ab
Summary of ANOVA /*t*-test (P values)							
Depth		0.030	< 0.01	< 0.01	0.104	< 0.01	< 0.01
Type of forest land		< 0.01	0.145	0.707	< 0.01	0.681	0.511
Depth × Type of forest land		< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01

Note: mean (±SE), n = 3. Entries in bold indicate the highest value between two types of forest land. Entries underlined indicate the highest value among three depths. Different capital letters A and B under same depth indicate significant differences ($P < 0.05$). Different lowercase letters a and b under same type of forest land indicate significant differences ($P < 0.05$). *BD*, Bulk density; *IWC*, initial soil water content; *TP*, total porosity; *CP*, capillary porosity; *NCP*, non-capillary porosity; *FWC*, field water capacity.

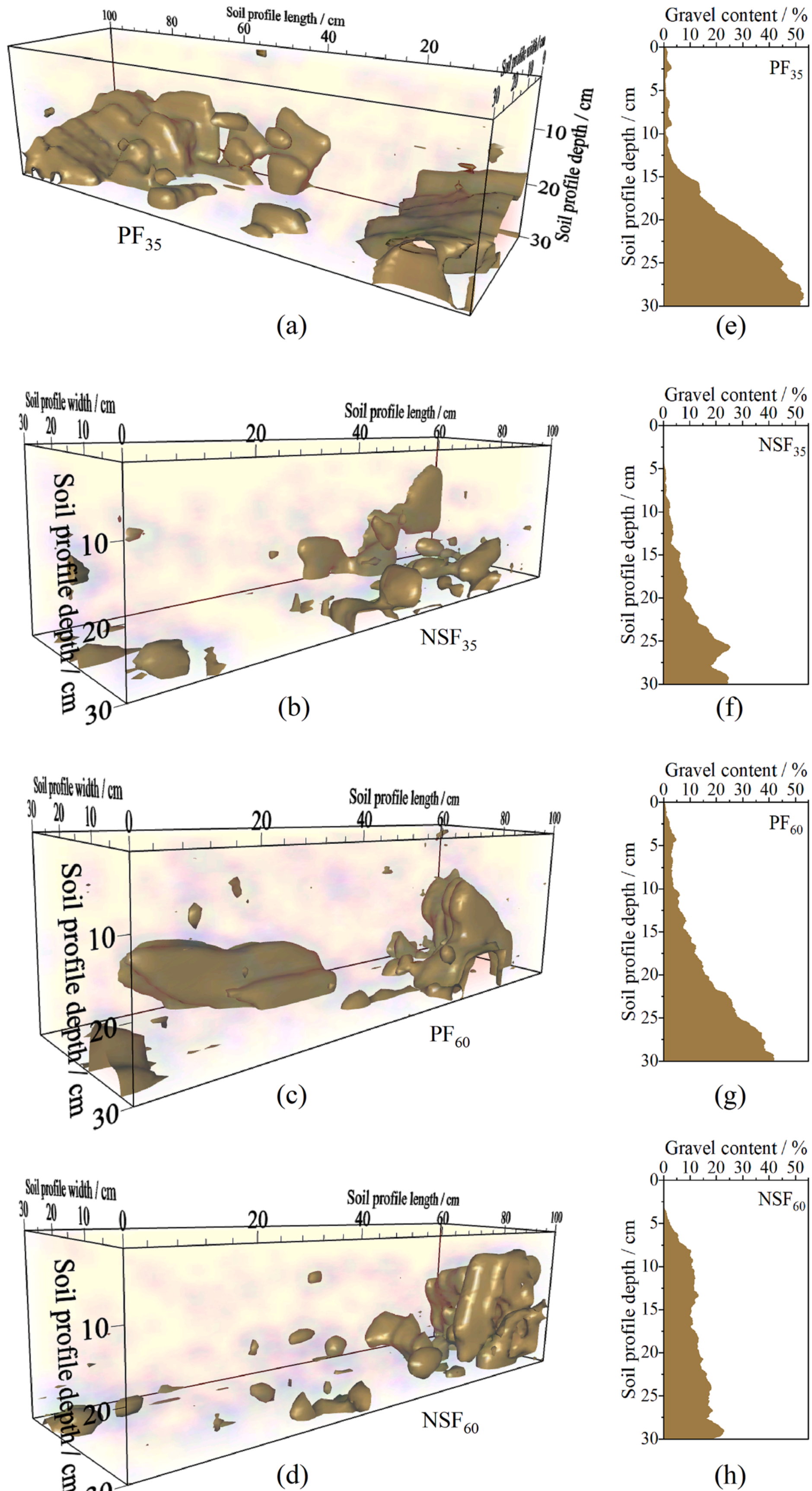

Fig. 4. Spatial distribution characteristics of gravel in the different reforestation sites. (a) and (b) spatial distribution of gravel in plantation forest (PF). (c) and (d) spatial distribution of gravel in natural secondary forest (NSF). (e)-(h) gravel content in each site corresponds to (a)-(d) respectively.

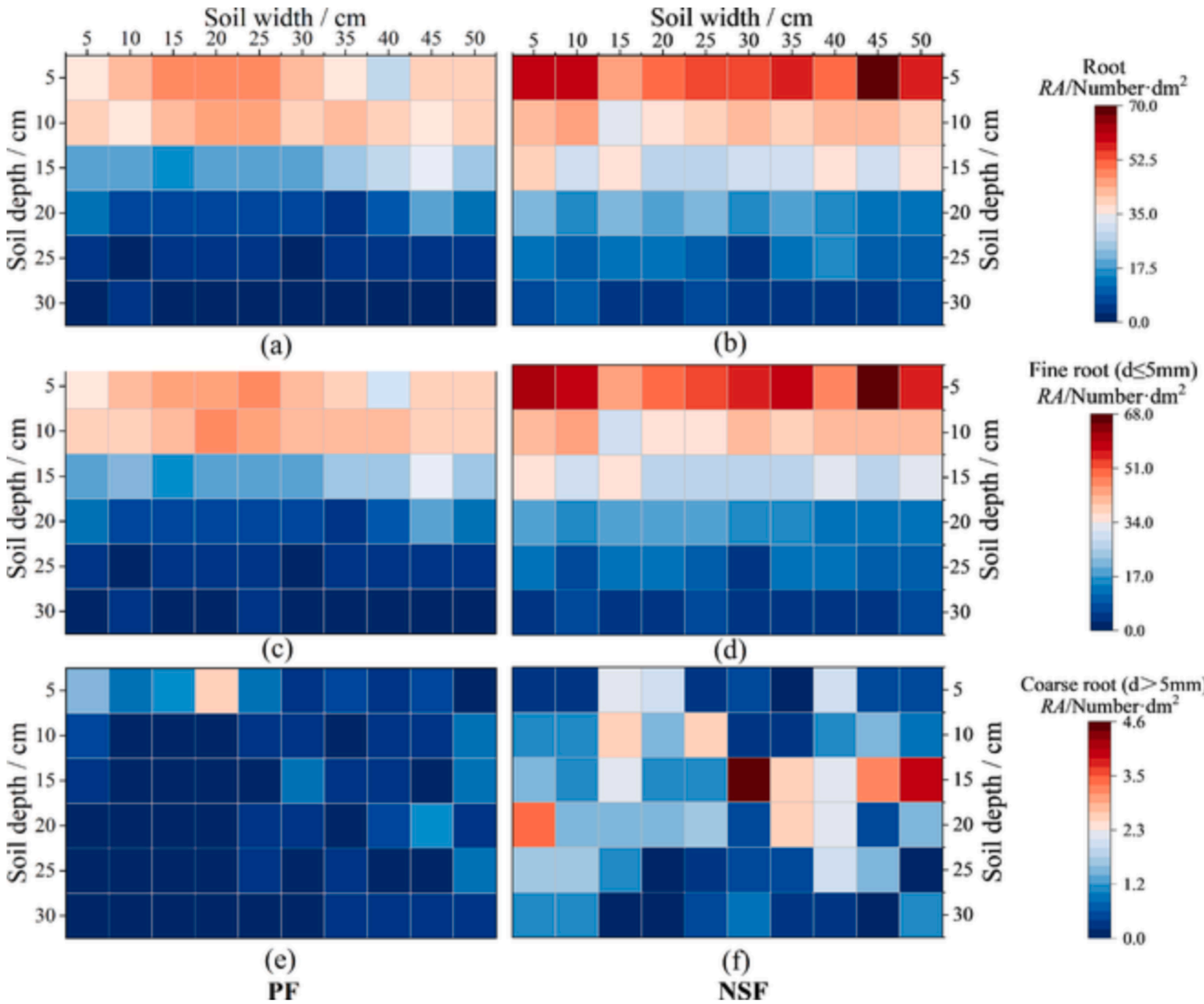

Fig. 5. Distribution of *RA* in the different reforestation areas. (a), (c), and (e) represent distribution of total roots, fine roots (d ≤ 5 mm), and coarse roots (d > 5 mm), respectively in plantation forest (PF). (b), (d), and (f) represent all root *RA*, fine root *RA*, coarse root *RA*, respectively, in natural secondary forest (NSF).

bottom, respectively, corresponding to a decrease in NSF of 82.8 % and 84.0 %. However, it is noteworthy that the distribution of coarse roots at each site did not show the same pattern. The PF had very few coarse roots in the 0–30 cm soil. The coarse roots of 0–10 cm (0.56 ± 1.75 dm^{-2}) in the PF site was significantly higher than that in 10–30 cm layer (0.22 ± 0.95 dm^{-2}) ($P < 0.01$). In contrast, the coarse roots in the NSF site was relatively more, mainly distributed within the 0–20 cm soil (76.9 %), of which the 10–20 cm layer (1.39 ± 2.95 dm^{-2}) was significantly higher than the 20–30 cm layer (0.74 ± 1.73 dm^{-2}) ($P < 0.01$).

In view of previous studies, fine roots play a more critical role in preferential flow movement (Cui et al., 2019). The distribution of fine roots in this study was indicative of the overall root level in the study area, as the proportion of fine roots exceeded 95 % (Fig. 5). For this reason, we further studied the spatial distribution characteristics of fine roots *RA* at the different sites (Fig. 6). It was obvious that the *RA* in the 0–5 cm layer of NSF was significantly higher than that of PF ($P < 0.01$), which was mainly affected by roots of surface herbs. The fine roots in PF in the 20–30 cm layer was significantly lower than that of NSF ($P < 0.01$), especially in PF_{35}. It is noteworthy that in combination with the distribution of the coarse roots, the coarse roots of the PF_{35} site were mainly distributed in the topsoil (0–10 cm), while the coarse roots of PF_{60} site were mainly distributed in the 10–30 cm layer. The roots of the PF_{35} site were more concentrated in the shallow layer, by either coarse or fine roots. In contrast, the roots of PF_{60} site had larger vertical downward extensibility. In addition, there was no significant difference in the distribution of *RA* between the two sites of NSF ($P > 0.05$), which was larger than that of PF and had tendency to extend to deeper soil layers.

3.2. Water infiltration

3.2.1. Saturated hydraulic conductivity (K_S)

We calculated K_S in the different reforestation areas (PF_{35}, NSF_{35}) and depending on dyed and undyed sections, for assessment of the infiltration capacity under different types of reforestation and infiltration as shown in Fig. 7. *Ks* showed a decreasing trend with increasing soil depth in all study areas, where the maximum value of K_S occurred in the natural secondary forest dyed section (NSF_D) 0–10 cm layer (1.05 ± 0.21 cm·min^{-1}) and the minimum in plantation forest undyed section (PF_UD) 20–30 cm layer (0.49 ± 0.16 cm·min^{-1}). For the same soil depth, K_S in NSF_D was significantly higher in the surface layer (0–10 cm) than the other areas ($P < 0.05$), related as NSF_D > plantation forest dyed area (PF_D) > natural secondary forest undyed area (NSF_UD) > PF_UD. In contrast, in the 10–30 cm layer, there was no significant difference in K_S among the four areas ($P > 0.05$). For the same area, dyed areas showed significant difference among different layers ($P < 0.05$). In PF_D, K_S decreased by 8.8 %, 46.0 % from 0–10 cm to 20–30 cm, and similarly, NSF_D decreased by 37.9 % and 27.8 % for the same depth. However, there was no significant difference among the three layers in the undyed areas ($P > 0.05$). Overall, under the same forest type conditions, the dyed K_S was higher than the corresponding undyed K_S, especially in the 0–20 cm layer.

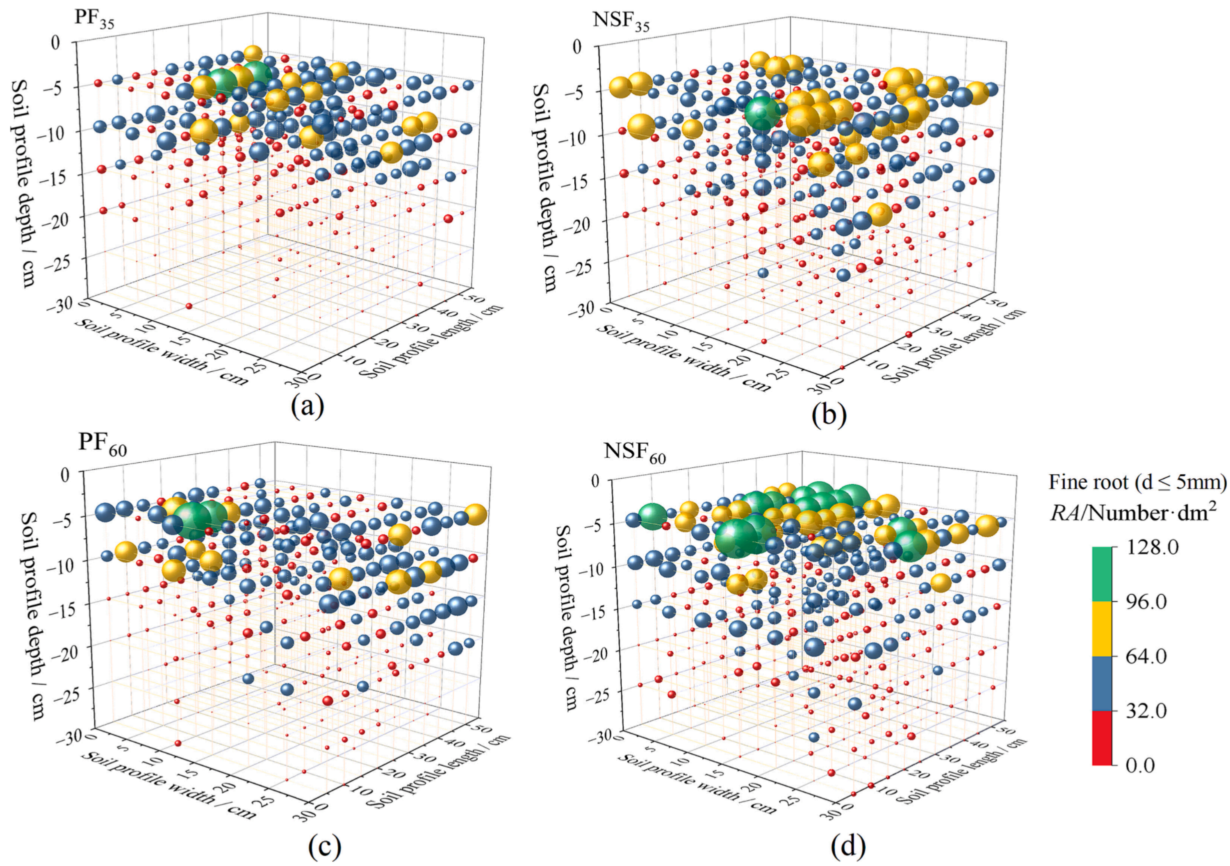

Fig. 6. Spatial distribution of fine roots *RA* in the different reforested areas. (a) and (c) spatial distribution of fine root *RA* in plantation forest (PF). (b) and (d) spatial distribution of fine roots *RA* in the natural secondary forest (NSF).

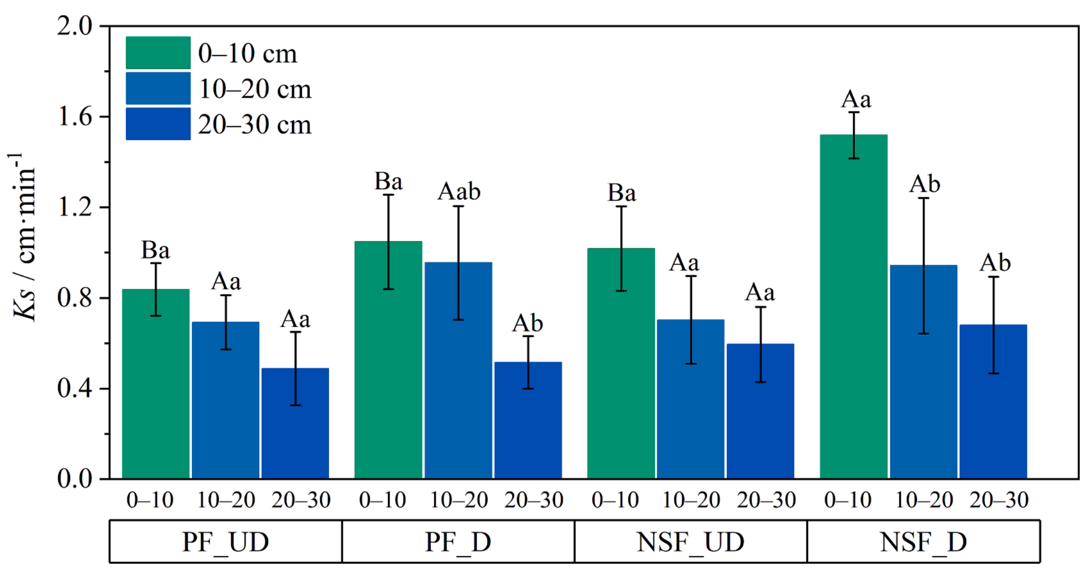

Fig. 7. Comparison of K_S (mean ± SE, n = 3) among different reforested types dyed areas and undyed areas. PF_UD, plantation forest undyed area. PF_D, plantation forest dyed area. NSF_UD, natural secondary forest undyed area. NSF_D, natural secondary forest dyed area. Bars with different capital letters (A and B) significant difference among different areas at the same soil depth at $P < 0.05$, and different lower-case letters (a and b) indicate significant difference among soil depth in the same area at $P < 0.05$.

3.2.2. Soil water content (SWC)

The distribution of *SWC* at different soil depths after infiltration at each site is shown in Fig. 8. There was a significant difference ($P < 0.01$) among the *SWC* of each layer at each site, as evidenced by a decreasing trend in *SWC* with increasing depth. Comparing *SWC* at different sites with the same depth, we found that, in the surface layer (0–10 cm), NSF was significantly higher ($P < 0.01$) than PF, with the largest *SWC* in NSF_{60} (36.1 ± 6.9 %) and the smallest in PF_{35} (32.2 ± 5.4 %). In the 10–20 cm soil layer the *SWC* was significantly higher for the extreme rainfall (60 mm) than for the heavy rainfall (35 mm) ($P < 0.01$).

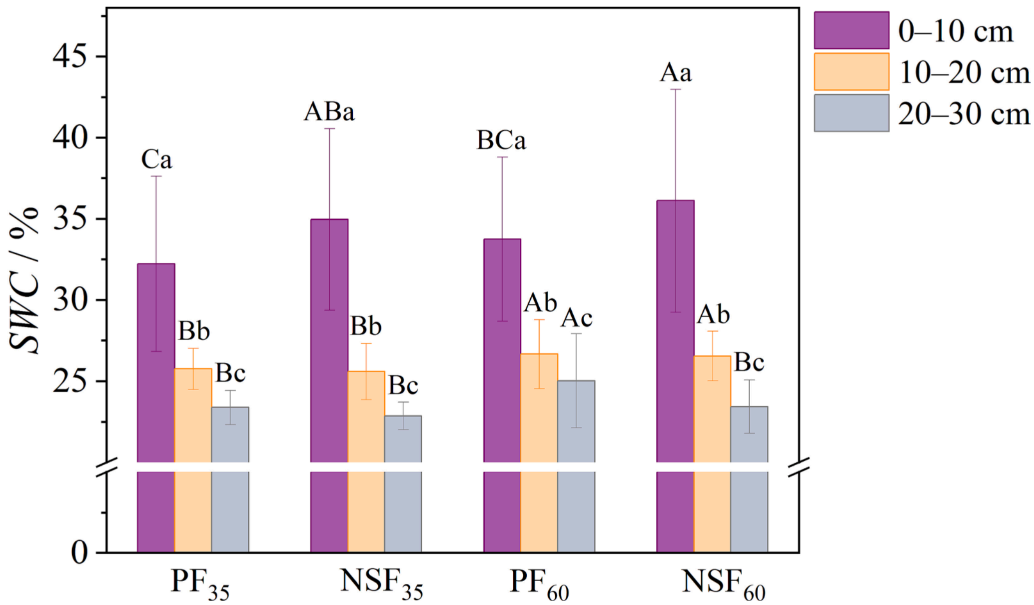

Fig. 8. Comparison of *SWC* (mean ± SE, n = 140) among different reforestation types. (a) and (b), PF_{35} and NSF_{35}, plantation forest and natural secondary forest under 35 mm infiltration amount, respectively. (c) and (d), PF_{60} and NSF_{60}, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively. Bars with different capital letters (A, B, and C) significant difference among different areas at the same soil depth at $P < 0.01$, and different lower-case letters (a, b, and c) indicate significant difference among soil depth in the same area at $P < 0.01$.

However, as the soil depth increased to 20–30 cm, the *SWC* of PF_{60} was significantly higher than that of the other sites ($P < 0.01$), as shown by PF_{60} (25.0 ± 2.9 %) > NSF_{60} (23.4 ± 1.6 %) > PF_{35} (23.4 ± 1.1 %) > NSF_{35} (22.7 ± 0.8 %).

The *IWC* is an index to characterize the distribution of *SWC* at the site before infiltration. Considering that *SWC* is a variable affected by external environment (i.e., season, rainfall, etc.), we further analyzed the difference between *SWC* and *IWC* (*SWC–IWC*) (Fig. 9) to better interpret the response of the spatial distribution of *SWC* to water infiltration at each site. We found that *SWC–IWC* < 0 and *SWC–IWC* ≥ 0 occurred at all sites, and increased infiltration increased the proportion of *SWC–IWC* ≥ 0. In particular, the proportion of *SWC–IWC* ≥ 0 was obviously higher for PF than NSF in the 0–10 cm layer, however, the opposite trend was shown in the 20–30 cm layer. The proportion of *SWC–IWC* ≥ 0 in PF decreased with the soil depth, while the opposite tendency was found for the NSF. This indicates that infiltration had a stronger effect on the increase of *SWC* in the PF surface layer, and a stronger increase in the 20–30 cm layer for NSF. Combined with the *IWC* data in Table 2, we found that the PF surface layer *IWC* was smaller than that for the NSF, while the 20–30 cm layer *IWC* was larger than for the NSF, suggesting that the increase in *SWC* is more likely to occur in areas with smaller *IWC*.

3.2.3. Spatial morphological characteristics of water flow paths

As there are seven vertically dyed profiles in each site, we selected a typical profile from each site for presentation (Fig. 10). Overall, all sites had a large area of dyeing within the top 5 cm, indicating a strong interaction between areas of dyed and undyed within the 0–5 cm soil layer. However, there were differences in vertically water flow path patterns between the PF and NSF. The water flow paths in PF were characterized by small number and large width, with an obvious shape of 'funnel'. And the NSF water flow paths were numerous and narrow, showing a more pronounced 'finger' pattern. As infiltration increase from heavy rainfall (35 mm) to extreme rainfall (60 mm), there was an increase in both horizontal and vertical connectivity. In addition, the water flow path isolated and disconnected with the surface appeared in each site.

The *DC* is a quantitative indicator of the extent to which water and solutes are transported through soil. Combining Fig. 10 and Fig. 11(a)-(d), the PF sites started to diverge sharply from about 2.5 cm depth, but the NSF sites started to diverge later at about 5 cm depth. In addition, we also found that *DC* decreased with the general soil depth, but the change was not completely the same, mainly due to the heterogeneity of soil spatial structure. From Fig. 11(a)-(d), we can see that the *DC* of NSF was slightly larger than that of PF under the same infiltration conditions, specifically as NSF_{35} (58.0 % ± 28.0 %) > PF_{35} (52.7 % ± 23.6 %), NSF_{60} (67.7 % ± 16.6 %) > PF_{60} (66.4 % ± 18.9 %). But combined with Fig. 12, there was no significant difference in *DC* between different reforestation types ($P > 0.05$). In addition, increasing the amount of infiltration resulted in a significant difference in *DC* for the same forest type ($P < 0.05$).

3.3. Relationships between infiltration parameters and soil characteristics

Soil physical properties, including *BD*, *TP*, *CP*, *NCP*, and *FWC*, were statistically significant correlated with infiltration parameters (Table 3 and Fig. 13). *BD* was significantly negatively correlated with infiltration parameters ($P < 0.01$), which could explain 65.5 % of K_S_D, 65.4 % of K_S_UD, 52.2 % of *SWC*, 50.2 % of *SWC–IWC*, 47.5 % of *DC* variation, respectively (Fig. 13(a), (f), (k)). *TP*, *CP*, *NCP*, and *FWC* were significantly positively correlated with K_S_D and K_S_UD ($P < 0.01$), which explained 76.3 %, 69.9 %, 44.4 %, 58.0 % of K_S_D, and 77.1 %, 64.2 %, 54.4 %, 56.5 % of K_S_UD variation, respectively (Fig. 13(b)-(e)). The results indicated that the reduction in *BD* increased the permeability of the soil, whereas the increase in *TP*, *CP*, *NCP*, and *FWC* enhanced the soil permeability. Note that *TP* contributed more to explain the variation in K_S. Similar to K_S, significantly positive correlation was found between *TP*, *CP*, *NCP*, *FWC* and *SWC*, *SWC–IWC*, and simultaneously explained 87.5 %, 92.5 %, 35.8 %, 58.8 % of the *SWC* variation, and 84.1 %, 88.7 %, 34.8 %, 54.4 % of the *SWC–IWC* variation. This indicated that increasing *TP*, *CP*, *NCP*, and *FWC* could enhance soil water retention capacity. Note that *CP* contributed more to explain the variation in *SWC* change than other soil properties (Fig. 13(g)-(j)). *DC* also showed significantly positive correlation with *TP*, *CP*, *NCP*, and *FWC* ($P < 0.05$), which explained 79.0 %, 88.0 %, 27.8 %, 70.2 % of *DC* variation, respectively (Fig. 13(l)-(o)). These findings show that the effects of soil

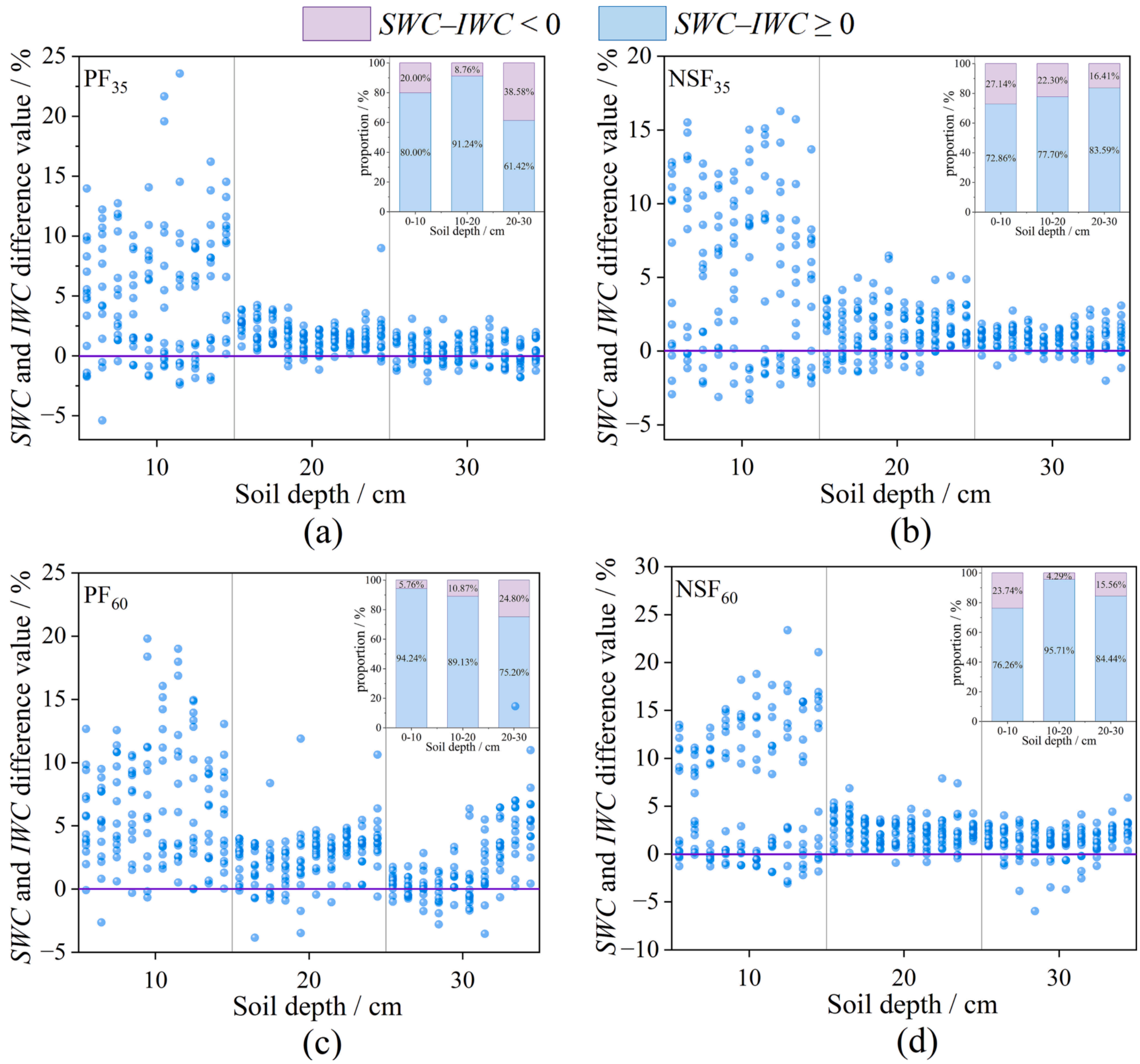

Fig. 9. Difference between soil water content (*SWC*) and initial soil water content (*IWC*) for different reforestation types. (a) and (b), PF_{35} and NSF_{35}, plantation forest and natural secondary forest under 35 mm infiltration amount, respectively. (c) and (d), PF_{60} and NSF_{60}, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively.

physical properties on soil plot scale infiltration are similar to the effects of soil physical properties on the infiltration capacity.

We excluded the gravel outcropping in the profile from *DC* calculations as water could not flow through the interior of gravel but only on its surface. However, relationships between *DC* and gravel did not display statistical significance. Therefore, we regard the dyed gravel as objects involved in water flow movement, and the undyed part as gravel that did not contribute to flow. For the 35 mm rainfall, more than 50 % of the gravel in the surface layer (0–10 cm) of both sites did not contribute to water flow transport (Fig. 14). The proportion of gravel involved in water flow in the 10–30 cm layer was different at the two sites. For PF, over 70 % of the gravel was involved in water flow movement, whereas only 40.5 % of the gravel in the NSF site contributed to water flow. As a result, under the condition of heavy rainfall (35 mm), gravel appears to have general effect on water infiltration. However, combined with the Fig. 4, it is observed that there exist certain disparities in the distribution of gravel morphology between PF_{35} and NSF_{35} sites. The former exhibits high gravel content and large volume, indicating good connectivity, whereas the latter displays scattered gravel with low connectivity. Therefore, gravel in PF_{35} can exert a more advantageous influence on water infiltration. By contrast, for the 60 mm rainfall, all layers of gravel had a positive effect on water flow. Therefore, increasing the rainfall will enhance the effects of gravel on water flow.

In terms of roots, we consider roots located in the dyed areas as involved in water transport. The proportion of fine roots involved in water transport tended to decrease with soil depth (Fig. 15), and NSF was significantly higher than PF ($P < 0.01$). Nearly all fine roots were involved in water transport in the surface layer 0–5 cm and there was a significant difference among different soil layers ($P < 0.01$). Additionally, as infiltration increased, a greater number of fine roots participated in the process of water infiltration. However, the distribution of coarse roots that may facilitate water infiltration exhibited no discernible pattern in PF sites (Fig. 15(a), (c)), but decreased with increasing soil depth in NSF sites (Fig. 15(b), (d)). This is primarily attributable to differences in the distribution of coarse roots resulting from variations in stand vegetation growth characteristics. Among the soil layers, the proportion of dyed coarse roots in PF and NSF sites was significantly

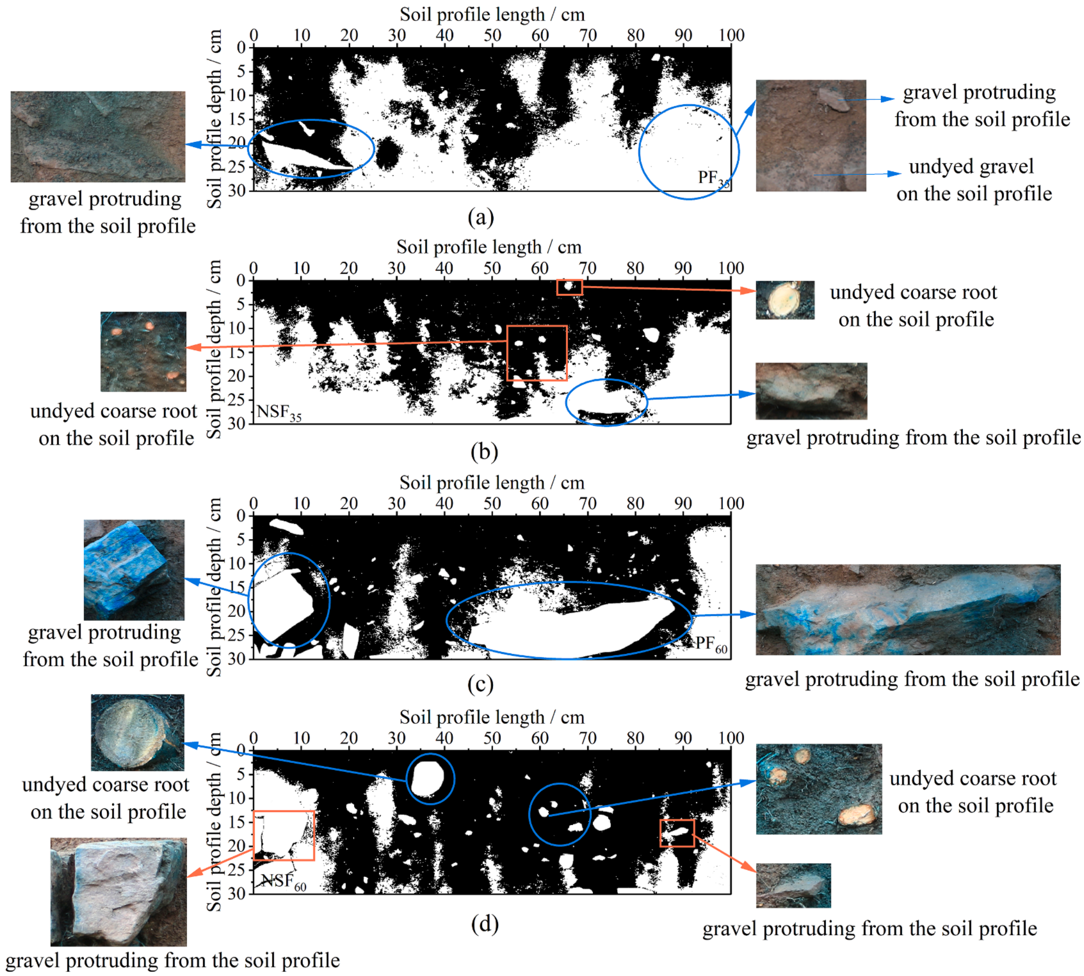

Fig. 10. Representative dyed soil profile sections for the different reforestation areas. (a) and (b), PF_{35} and NSF_{35}, plantation forest and natural secondary forest under 35 mm infiltration, respectively. (c) and (d), PF_{60} and NSF_{60}, plantation forest and natural secondary forest under 60 mm infiltration, respectively.

different ($P < 0.05$). However, coarse roots only accounted for 2.6 % of the overall dyed roots and were absent in certain soil layers, leading to significant variations in the number of dyed coarse roots for the two forest types. Therefore, fine roots played a predominant role in promoting water infiltration within the study area. For this reason, we conducted further analysis on the correlation between fine root *RA* and its corresponding *DC* in the smallest unit within the core area of the profile, aiming to gain a better understanding of how roots promote water infiltration. As can be seen from Fig. 16, there was a significant positive correlation between fine root *RA* and *DC* ($P < 0.01$), and the Pearson correlation coefficient showed that the infiltration amount of 35 mm ($r = 0.377$) was greater than that of 60 mm ($r = 0.294$), which explain 14.2 % and 8.6 % of *DC* variation, respectively.

4. Discussion

4.1. Difference in soils for different reforestation types

Reforestation is widely recognized as an essential factor affecting soil characteristics. Poelking et al. (2015) found that soil physiochemical characteristics were related to vegetation types and coverage in the study of periglacial landform of the Porter Peninsula in the Antarctic Marine Climate Zone. Similarly, our study found that reforestation can significantly improve soil physical properties and *RA*, and that physical properties (Table 2) and *RA* (Fig. 5) varied significantly with soil depth. Higher *BD*, lower *TP*, *CP*, *NCP*, *IWC*, *FWC*, and *RA* occurred in deeper soil compared to shallow soil. The distribution of roots and the activity of animals and microorganisms in shallow soil can increase porosity and decrease *BD* (Chen et al., 2018). In addition, shallow soil is more susceptible to external forces, such as human and mechanical stampede compaction (Liu et al., 2016). The differences between *TP*, *CP*, and *NCP* of PF and NSF became smaller with increasing soil depth (Table 2), which also proves this point. Capillary pores have capillary action, which can not only transport water but also retain water, water in capillary pores can provide water for vegetation growth. Thus, *CP* is an index reflecting soil water retention capacity. Water in non-capillary pores can transport water, which is the channel of water and air, and can effectively reduce surface runoff, thus, *NCP* is an indicator of soil permeability. We found that with the increase of soil depth, the water retention and permeability of soil decreased to varying degree, which is consistent with the previous research (Wang et al., 2023). Regarding the distribution of gravel, it showed an increasing trend with soil depth (Fig. 4), as well as a marked spatial heterogeneity, with slight differences in gravel content at each site. It is worth noting that, considering that the study area is in a paleo-periglacial landform area, surface soils in the 0–30 cm were distributed with a high gravel content of varying volume, an aspect that differs considerably from the soil characteristics of other landform types.

Under different reforestation approaches in the paleo-periglacial landform of eastern Liaoning mountainous regions, higher *BD*, lower *TP*, *NCP*, and *FWC* occurred in PF compared to NSF (Table 2), mainly due to stronger compaction of PF surface soil by anthropogenic activities. The root distribution characteristics of different plants are varied, and root channels can change the porosity, aggregate stability, and

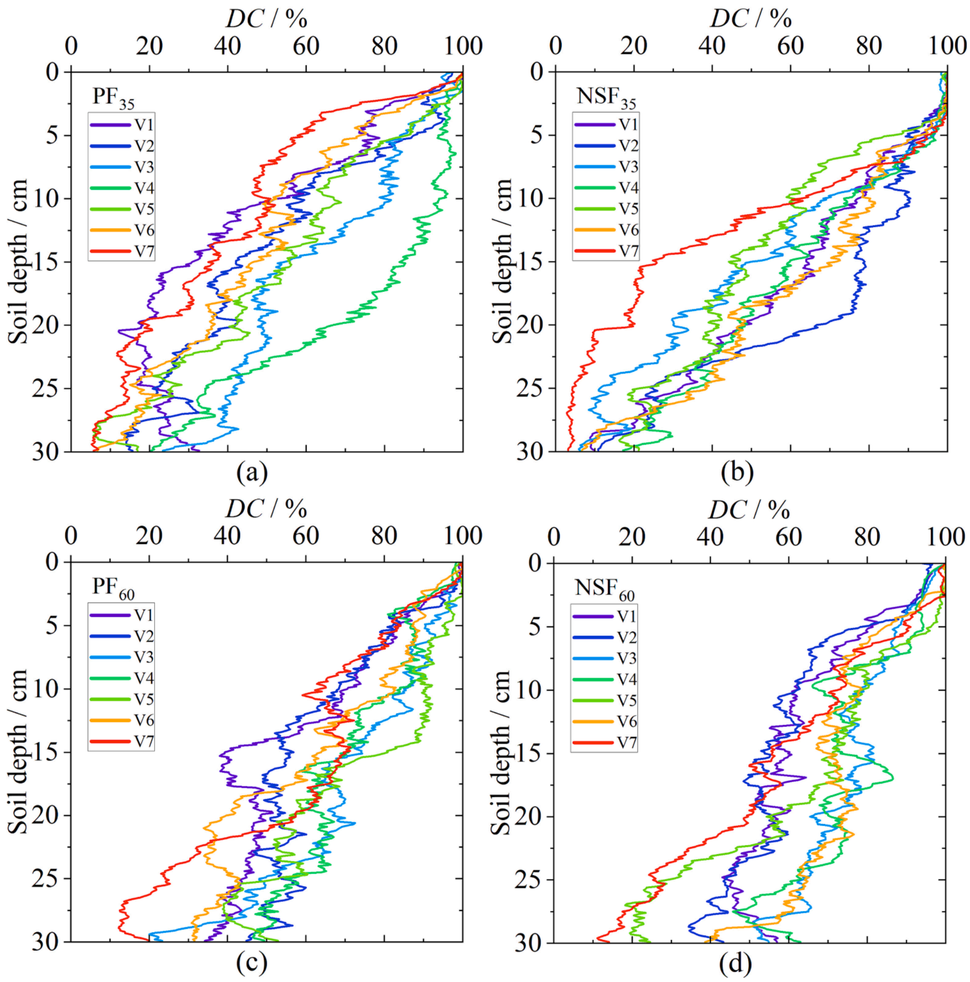

Fig. 11. Variation of dyed area with soil depth for the different reforestation types and different infiltration amount. (a) and (b), PF_{35} and NSF_{35}, plantation forest and natural secondary forest under 35 mm infiltration, respectively. (c) and (d), PF_{60} and NSF_{60}, plantation forest and natural secondary forest under 60 mm infiltration, respectively. V1-V7 represent the eight profiles at each site.

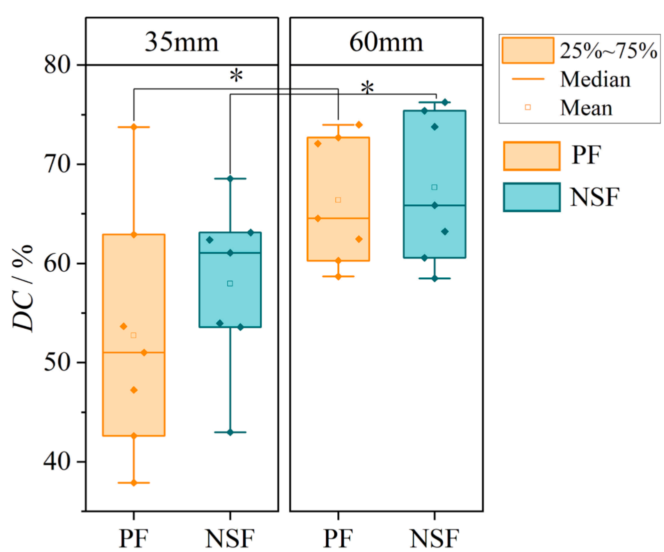

Fig. 12. Boxplots (mean ± SE, n = 7) of the dye coverage ratio (*DC*) for the plantation forest (PF) and natural secondary forest (NSF) under 35 mm and 60 mm infiltration. Boxes with * indicate significant difference at $P < 0.05$.

Table 3
Correlation coefficients between soil physical properties and infiltration parameters.

Infiltration parameter	*BD*	*TP*	*CP*	*NCP*	*FWC*
*Ks*_D	−0.809***	0.783***	0.687***	0.666**	0.762***
*Ks*_UD	−0.809***	0.828***	0.705***	0.737***	0.751***
SWC	−0.722**	0.871***	0.860***	0.598**	0.767***
SWC–IWC	−0.708**	0.793***	0.753***	0.590**	0.737**
DC	−0.689**	0.834***	0.854***	0.528*	0.838***

Note: K_S_D, saturated hydraulic conductivity in dyed area; K_S_UD, saturated hydraulic conductivity in undyed area; *SWC*, soil water content; *SWC–IWC*, difference value of soil water content and initial soil water content; *DC*, dye coverage; *BD*, bulk density; *TP*, total porosity; *CP*, capillary porosity; *NCP*, non-capillary porosity; *FWC*, field water capacity.
* $P < 0.05$; ** $P < 0.01$, *** $P < 0.001$.

organic matter of nearby soil, thus enhancing soil infiltration (Chandler and Chappell, 2008). We found that, in surface soil, *NSF* was higher than PF in both fine root *RA* and coarse root *RA*. The difference in root distribution led to an increase in soil porosity and a decrease in *BD* for NSF. Among them, the distribution of fine roots in the surface soil, especially in the 0–5 cm layer, reflects the roots of herbs. It is generally believed that NSF has richer herb types, higher coverage, and more vigorous roots. The deeper fine root distribution was higher for NSF than PF, which was mainly the fine roots of trees. According to Fig. 4(a)-(d) and

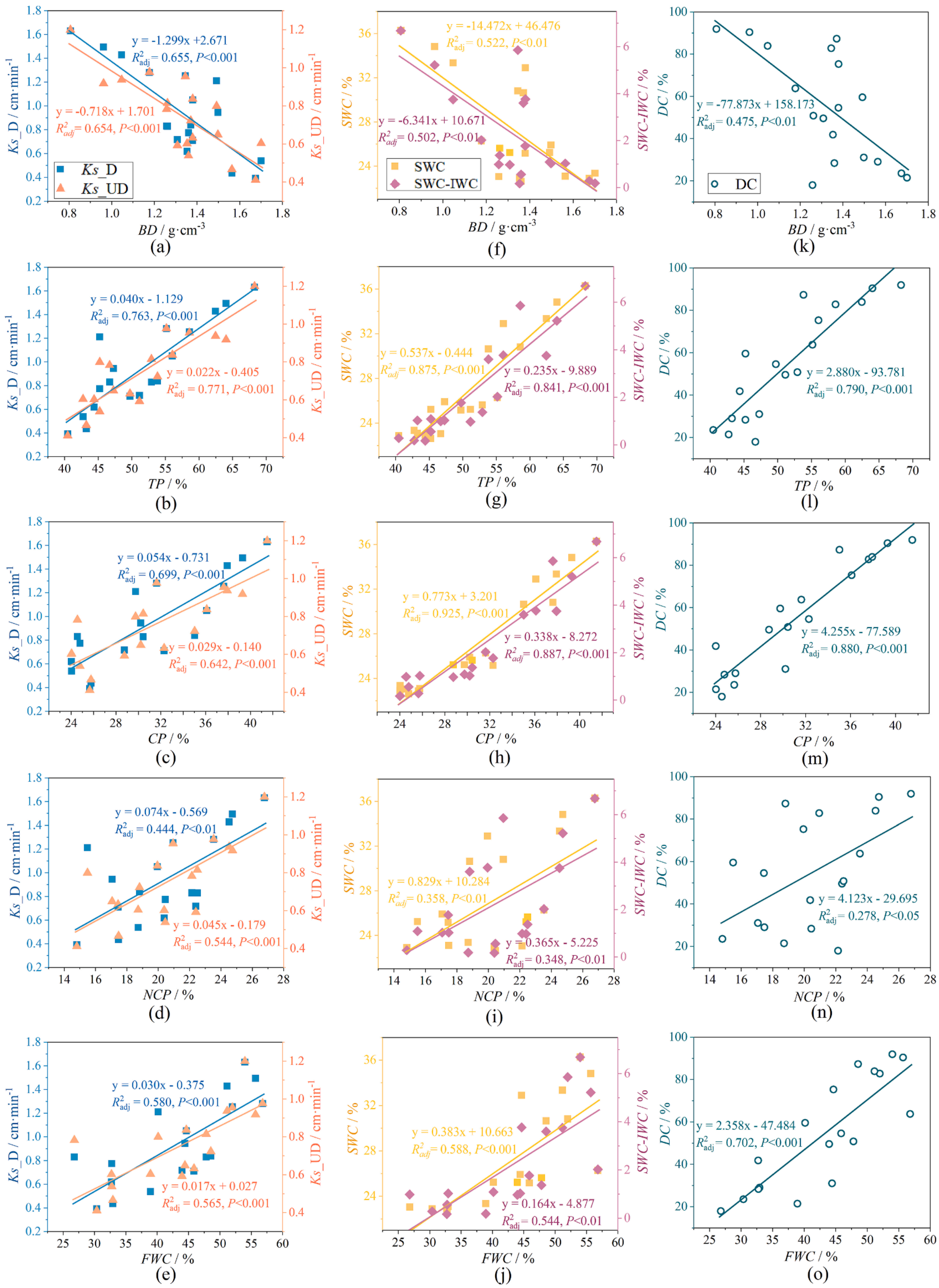

Fig. 13. Relationship between soil physical properties (i.e., *BD*, *TP*, *CP*, A*NCP* and *FWC*) and infiltration parameters (i.e., K_S_D, K_S_UD, *DC*, *SWC* and *SWC–IWC*). *BD*, bulk density; *TP*, total porosity; *CP*, capillary porosity; *NCP*, non-capillary porosity; *FWC*, field water capacity; K_S_D, saturated hydraulic conductivity in dyed area; K_S_UD, saturated hydraulic conductivity in undyed area; *DC*, dye coverage ratio; *SWC*, soil water content; *SWC–IWC*, difference value of soil water content and initial soil water content. The regression analysis was only presented if it was significant at $P < 0.05$.

gravel distribution, the large size of gravel in the PF plot limited the growth space of roots. the distribution of coarse roots represented the distribution of tree roots in PF and NSF forests, the coarse roots in NSF site were more distributed in the surface layer (Fig. 5), in view of the special geomorphic types and different reforestation approaches in the study area, PF is artificially planted and managed, showing better vegetation stability, but the NSF vegetation is less stable, which we found during the sample site survey where the coarse roots of the NSF site was exposed to the ground surface. This is mainly due to the low *BD* and high *TP* of NSF, which is more prone to soil erosion after high-intensive rainfall. Therefore, future management of the NSF should focus on strengthening the soil erosion control aspects to enhance vegetation stability. In contrast, PF management should consider the sustainable development of the forest land. Overall, the gravel content

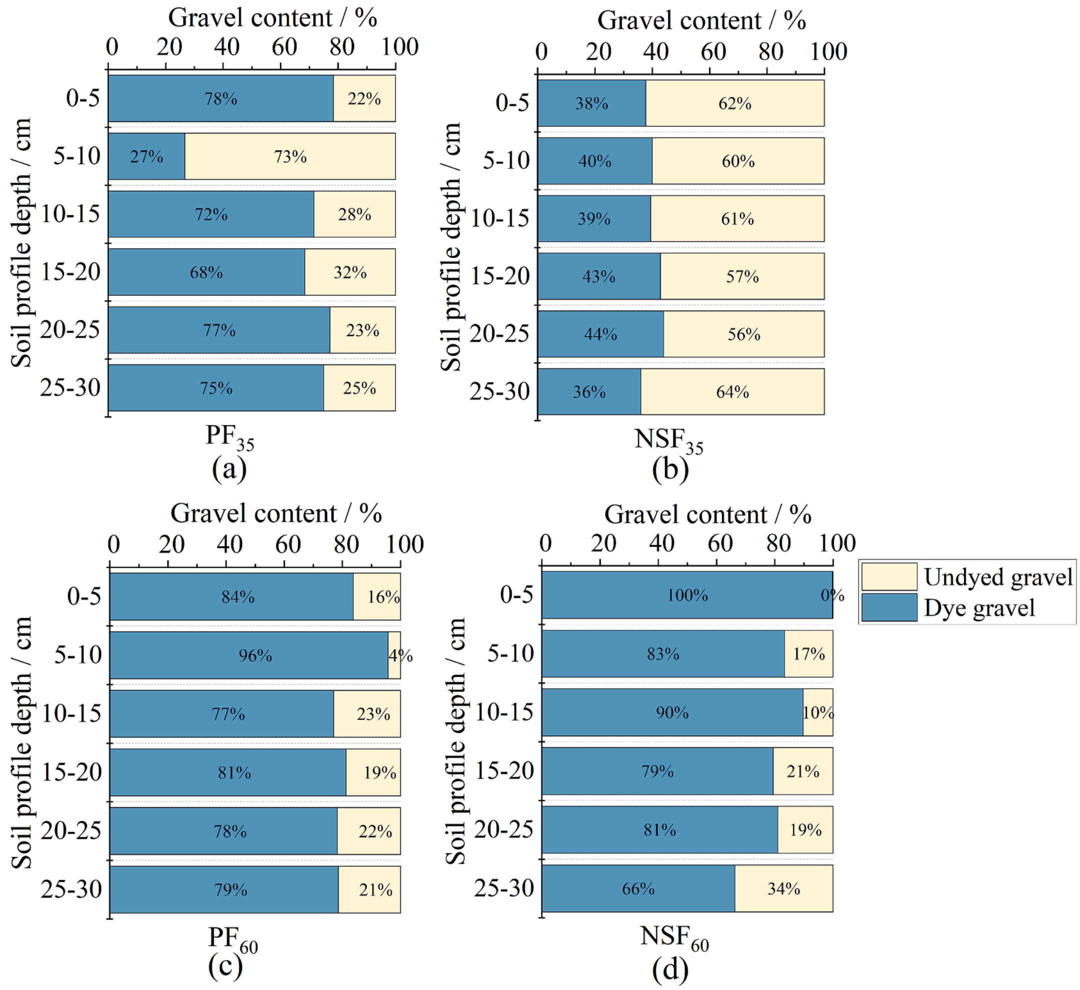

Fig. 14. Proportion of dyed and undyed gravel at different soil depths for the different reforestation types. (a) and (b), PF_{35} and NSF_{35}, plantation forest and natural secondary forest under 35 mm infiltration amount, respectively. (c) and (d), PF_{60} and NSF_{60}, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively.

was higher in PF than in NSF, especially at 15 cm depth where there was a sudden increase (Fig. 4(e)-(h)), and it showed randomness as well. In addition, the gravel in PF exhibited greater volume and connectivity, while those in the NSF were more fragmented and less connected.

4.2. Effect of reforestation on soil infiltrability

On the plot scale, we explored the water infiltration paths (Fig. 10). We found that there is an obvious preferential flow phenomenon in the study area, and the preferential flow is the main pattern of water infiltration in the profile. Preferential flow, is a phenomenon whereby water and solutes bypass most of the soil matrix and infiltrate rapidly into the deeper soil or groundwater layers along pathways such as pores, fissures, wormholes, and joints, characterized by spatial heterogeneity, instability, imbalance, encirclement, and rapidity (Beven and Germann, 1982; Ritsema and Dekker, 2000; Zhang et al., 2016a). Preferential flow has an important impact on soil infiltration, which can increase the complexity of soil water redistribution patterns. We found that preferential flow patterns differed between the different reforestation approaches, with 'funnel' shaped flow occurring in PF and 'finger' shaped flow in NSF. As could be seen, the lateral connectivity of the flow paths was larger in PF, resulting in a wider 'funnel' flow than for NSF. The infiltration of heterogeneous soil tends to form unstable wetting fronts and may break into narrow wetting columns or 'finger' shape or 'funnel' shape flow due to soil structural characteristics (Kim et al., 2005). The activities of animals, plants and microorganisms in soil will have a positive effect on the formation of unstable wetting front (Morales et al., 2010), which in turn affect water infiltration.

Overall, the *DC* of NSF was slightly higher than that of PF, but there was no significant difference (Figs. 11-12). In addition, when the rainfall increased from heavy rainfall (35 mm) to extreme rainfall (60 mm), the *DC* increased, and the hydrological connectivity was enhanced (Fig. 10). Correlation analysis showed that *DC* was negatively correlated with *BD* and positively correlated with *TP*, *CP*, *NCP*, *FWC*, and fine roots *RA* (Table 3 ; Fig. 13 ; Fig. 16), which strengthens our hypothesis. Bogner et al. (2010) found an effect of *BD* on preferential flow in Norway Spruce forest and highlighted the role of roots in promoting preferential flow. In this study, the NSF soil surface layer had higher *RA*, *TP*, and *DC*, which showed that the pore channels formed by the well-developed root system are more conducive to soil infiltration. Besides, due to the low number of coarse roots in this study, there was no regular effect on water infiltration. The role of fine roots in water infiltration decreased with soil depth, and higher rainfall led to increased contribution of fine roots to infiltration (Fig. 15). Fine roots were found to be the primary type influencing infiltration in the study region. The gravel distribution began to increase quickly at around 15 cm depth and the dyed gravel appeared random in the profile (Fig. 10). In contrast to continuous pore channels formed by roots extending vertically or laterally from the surface (Bargués Tobella et al., 2014), pore channels formed by gravel and soil are more random. Previous studies have shown that due to the

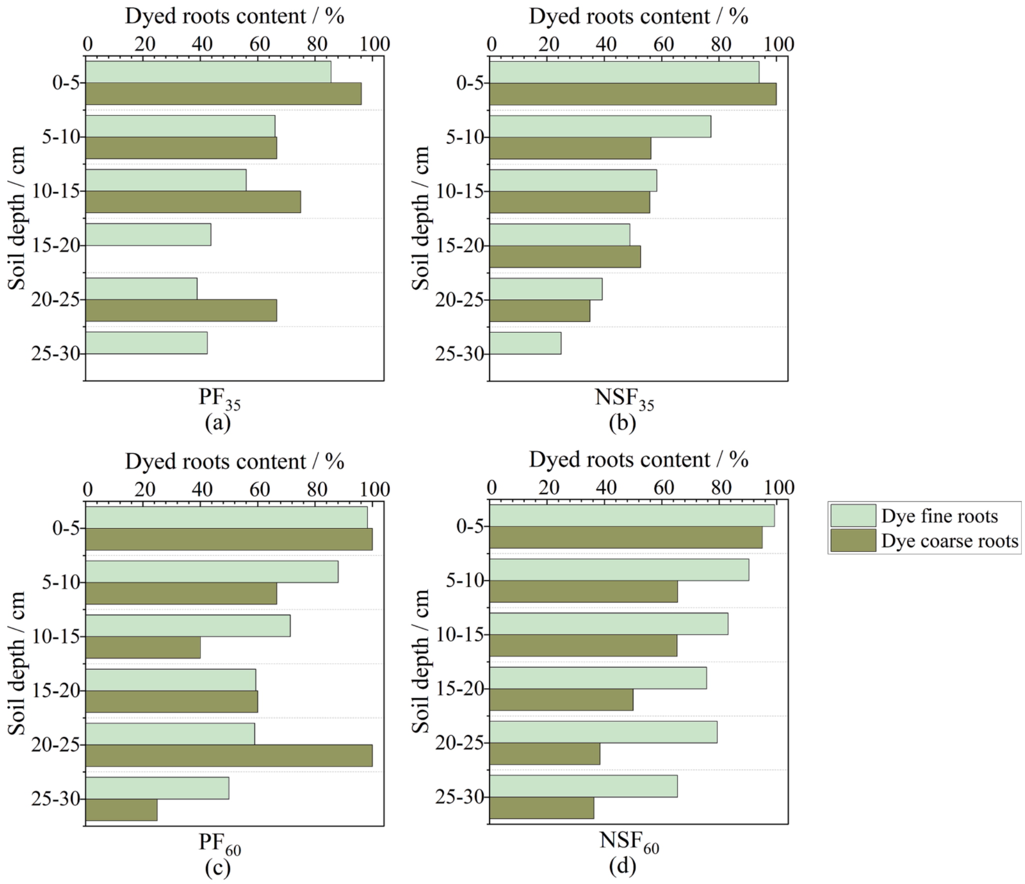

Fig. 15. Proportion of dyed fine roots and dyed coarse roots in different soil depths in the reforested areas. (a) and (b), PF_{35} and NSF_{35}, plantation forest and natural secondary forest under 35 mm infiltration amount, respectively. (c) and (d), PF_{60} and NSF_{60}, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively.

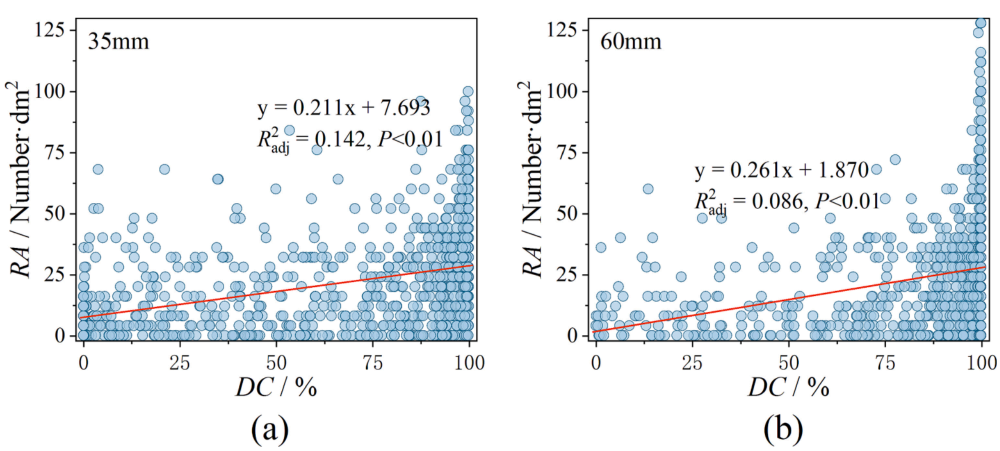

Fig. 16. The relationships between fine roots *RA* and *DC* under different infiltration amount. (a) forests under 35 mm infiltration amount, (b) forests under 60 mm infiltration amount.

difference of specific heat capacity between gravel and soil, affected by the change of external environment (i.e., dry-wet alternation, freeze–thaw alternation, etc.), there will be cracks between them (Gargiulo et al., 2015), which could increase the flow path (Zhang et al., 2016b). However, our investigation into the three-dimensional arrangement of gravel (Fig. 4) revealed that, in terms of enhancing water infiltration, larger volumes of gravel are more effective when they are aggregated as opposed to being scattered and small. These findings are consistent with previous studies which concluded that roots, gravel, and soil properties are the main factors influencing water infiltration (Luo et al., 2022; Mei et al., 2018; Qi et al., 2020). In addition, our findings indicate that as infiltration increased from heavy to extreme, the contribution of both gravel and roots to infiltration increased, due to an increase in hydrological connectivity as rainfall increased, with more water channels involved in infiltration, most of which was due to the macropore network formed between roots and gravel and soil, this finding confirms our suspicion. However, with low connectivity of the macropore network, it is evident from Fig. 10, Fig. 14, and Fig. 15 that not all roots and gravel promote water infiltration.

Soil infiltration contributes to the reduction of surface runoff and is an important indicator of soil erosion resistance (Ellison, 1945). Different reforestation approaches are generally considered to influence infiltration characteristics by changing the physicochemical properties of the soil (Sun et al.,2018; Qiu et al., 2023). We analyzed the soil infiltration capacity of the preferential flow area (dyed area) and the matrix flow area (undyed area) of different reforestation forests by measuring K_S at the cutting cylinder scale. Generally, K_S decreased with the increase of soil depth. In the surface layer (0–10 cm), NSF K_S was higher than PF, while the difference in K_S between the two sites in the 10–30 cm soil layer was not significant (Fig. 7). As we expected, this result showed that the different reforestation approaches in the study area significantly altered the infiltration capacity of the surface layer soil (0–10 cm). The surface (0–10 cm) of PF was more affected than the 10–30 cm layer, thus the K_S of PF deep layer (10–30 cm) was slightly different from NSF. Table 3 and Fig. 13 showed that soil physical properties (*BD*, *TP*, *CP*, *NCP*, and *FWC*) played a decisive role in soil infiltration, which is consistent with our conjecture. As can be seen, the increase in porosity has a positive effect on infiltration (Wang et al., 2023). These results are consistent with the findings of previous studies on the effects of water infiltration in forest lands (Qiu et al., 2023; Wang et al., 2023) and plantations (Zhang et al., 2022). Different reforestation approaches change the soil structure and physical properties, which are mainly influenced by the organic cementing substances and biomass in soil (Nair et al., 2010). In addition, K_S was higher in the preferential flow area than in the matrix flow area for each site, and there was a significant difference in K_S between soil layers in the area where preferential flow occurred, while there was no significant difference in K_S between layers in the matrix flow area (Fig. 7), indicating that preferential flow was the main factor contributing to the difference in water infiltration in different site profiles.

We investigated and analyzed the *SWC* at the centimeter scale on the soil profile after infiltration and found that infiltration could change the redistribution of *SWC* (Fig. 8), especially for the 0–10 cm layer *SWC*, which was consistent with the previous study conclusions (Sheng et al., 2014; Luo et al., 2022). Overall, *SWC* decreased with the increase of soil depth. The *SWC* of NSF was higher than that of PF in 0–10 cm layer, but PF was higher than NSF with the soil depth, which indicated that PF may show better water retention capacity with the increase of soil depth. Furthermore, we characterized the moisture trends at each site by *SWC–IWC* values (Fig. 9), and the results showed that both $SWC–IWC < 0$ and $SWC–IWC \geq 0$ occurred in each site, which is similar to the results of Luo et al. (2022). One of the reasons we analyze is that the distribution of *SWC* is spatially heterogeneous. Secondly, there is immobile water in soil. Thirdly, the roots in soil absorb water. Fourthly, the possibility of "new water" carrying away "old water" to deep soil after infiltration, given the rapid transport characteristics of preferential flow. Correlation analysis showed that both *SWC* and *SWC–IWC* were negatively correlated with *BD* and positively correlated with *TP*, *CP*, *NCP*, *FWC* (Table 3; Fig. 13). *CP* is an important indicator of water retention capacity of the soil, and our findings showed that *CP* has the strongest explanatory for *SWC* and *SWC–IWC*, which are 92.5 % and 88.7 %, respectively. Combined with the changes of *CP* with soil depth in different reforestation approaches in Table 2, it was further shown that PF has better water retention capacity in deeper soil (10–30 cm) than NSF, which also confirms our conjecture. In addition, combined with the *IWC* distribution of each site, the smaller the *IWC*, the greater the probability of $SWC–IWC \geq 0$, meaning that the increase in *SWC* after infiltration tends to occur in areas with low *IWC*. This may be due to the positive effect of *IWC* on soil water potential and the negative effect on soil water absorption capacity (Liu et al.,2019).

5. Conclusions

We quantified differences and change patterns in soil water infiltration under different reforestation approaches and their response to soil characteristics in a typical paleo-periglacial landform of eastern Liaoning mountainous regions, China. The main conclusions are as follows:

(1) The phenomenon of preferential flow appeared at each site in the study area, and preferential flow was the main infiltration type at the plot scale, which was also the main factor contributing to the difference in water infiltration in different site profiles. The flow path pattern differed between PF and NSF, with PF exhibiting 'funnel' shape and NSF 'finger' shape, and increased infiltration increased the connectivity of the flow paths.
(2) The results of *Ks* and *DC* indicated that NSF exhibited slightly higher soil infiltration capacity than PF, but there was no significant difference ($P > 0.05$). The infiltration capacity of the soil increased significantly when infiltration increased from heavy rainfall (35 mm) to extreme rainfall (60 mm), especially in the 0–10 cm layer. In addition, under the same forest type conditions, the *Ks* of the preferential flow areas was higher.
(3) The redistribution of water infiltration by the different reforestation approaches showed differences, specifically in the increase of *SWC* in the surface 10 cm of NSF and 20–30 cm of PF, mainly due to the differences in soil characteristics between PF and NSF. Furthermore, the increase in *SWC* was more likely to occur in areas with lower initial water content.
(4) The PF had higher *BD* and smaller *TP*, *CP*, *NCP*, *IWC*, and *RA* in the 0–20 cm layer, but there was little difference in the distribution of gravel, which in turn resulted in NSF having a better infiltration capacity than PF in 0–20 cm. In the 20–30 cm layer, *CP* and *IWC* were higher for PF than in NSF, indicating a higher water retention capacity in PF.
(5) Infiltration capacity parameters (including *Ks*, *SWC*, *SWC–IWC*, and *DC*) were significantly correlated with *BD*, *TP*, *CP*, *NCP*, *FWC* and fine roots *RA* ($P < 0.05$). The larger the volume of gravel in the profile lead to larger connectivity and facilitation of water infiltration, compared to fragmented gravel. In addition, fine roots were the main root type in promoting water infiltration.

Our results demonstrate that properly designing PF could ameliorate soil water infiltration patterns by altering soil characteristics, thereby influencing soil water redistribution processes. Generally, NSF has a higher infiltration and water retention capacity. However, with increasing soil depth to the 20–30 cm layer, PF has a slightly stronger retention capacity than NSF. The PF can reduce runoff and soil erosion as well as NSF, whereas in terms of vegetation stability and deep soil water retention, PF is more advantageous. Therefore, it is important to establish the PF in a rational manner and to enhance the protection of the stability of the vegetation in the NSF in future in the study area, with a view to better conserve water resources and prevent soil and water loss. These findings have important implications for a better understanding of the influence of forestland in paleo-periglacial landform of eastern Liaoning mountainous regions and other periglacial landform area on soil hydrological processes, and have profound implications for reforestation and the establishment and management of forest ecosystem and water conservation forest in these areas.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was funded by the National Natural Science Foundation of China (41877154), the National Key Research and Development Program of China (2022YFF1300804) and the National Natural Science Foundation of China (41977074).

References

Alaoui, A., Caduff, U., Gerke, H., Gerkeb, H., Weingartnera, R., 2011. A preferential flow effects on infiltration and runoff in grassland and forest soils. Vadose Zone J. 10, 367–377. https://doi.org/10.2136/vzj2010.0076.

Bargués Tobella, A., Reese, H., Almaw, A., Bayala, J., Malmer, A., Laudon, H., Ilstedt, U., 2014. The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. Water Resour. Res. 50 (4), 3342–3354. https://doi.org/10.1016/j.geoderma.2019.113978.

Barral, M.P., Rey Benayas, J.M., Meli, P., Maceira, N.O., 2015. Quantifying the impacts of ecological restoration on biodiversity and ecosystem services in agroecosystems: a global meta-analysis. Agr Ecosyst Environ 202, 223–231. https://doi.org/10.1016/j.agee.2015.01.009.

Beven, K., Germann, P., 1982. Macropores and Water Flow in Soils. Water Resour. Res. 1982 (18), 1311–1325. https://doi.org/10.1029/WR018i005p01311.

Bogner, C., Gaul, D., Kolb, A., Schmiedinger, I., Huwe, B., 2010. Investigating flow mechanisms in a forest soil by mixed-effects modelling. Eur. J. Soil Sci. 61 (6), 1079–1090. https://doi.org/10.1111/j.1365-2389.2010.01300.x.

Brid, N.R.H., Bartoli, F., Dexter, A.R., 1996. Water retention models for fractal soil structures. Eur. J. Soil Sci. 47 (1), 1–6. https://doi.org/10.1111/j.1365-2389.1996.tb01365.x.

Chandler, K.R., Chappell, N.A., 2008. Influence of individual oak (Quercus robur) trees on saturated hydraulic conductivity. For. Ecol. Manag. 256 (5), 1222–1229. https://doi.org/10.1016/j.foreco.2008.06.033.

Chen, C.F., Liu, W.J., Wu, J.N., Jiang, X.J., 2018. Spatio-temporal variations of carbon and nitrogen in biogenic structures of two fungus-growing termites (M. annandalei and O. yunnanensis) in the Xishuangbanna region. Soil Biol. Biochem. 117, 125–134. https://doi.org/10.1016/j.soilbio.2017.11.018.

Chen, C.F., Zou, X., Singh, A.K., Zhu, X.A., Zhang, W.J., Yang, B., Jiang, X.J., Liu, W.J., 2021. Effects of hillslope position on soil water infiltration and preferential flow in tropical forest in southwest China. J. Environ. Manage. 299, 113672 https://doi.org/10.1016/j.jenvman.2021.113672.

Cui, Z., Wu, G.L., Huang, Z., Liu, Y., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. J. Hydrol. 578, 124023 https://doi.org/10.1016/j.jhydrol. 2019.124023.

Cui, Z.J., Zhu, C., 1988. Commentaries and prospects of the study on periglacial geomorphology. J. Glaciol. Geocryol. 10 (3), 304–311.

Danielson, R.E, Sutherland, PL., 1986. Methods of Soil Analysis: Part 1 Physical and Mineralogical Methods. Klute, A., Eds.; Madison, WI: American Society of Agronomy, Soil Science Society of America. pp. 443-461.

Demir, G., Michalzik, B., Filipzik, J., Metzger, J.C., Hildebrandt, A., 2022. Spatial variation of grassland canopy affects soil wetting patterns and preferential flow. Hydrol Process 36 (12), e14760. https://doi.org/10.1002/hyp.14760.

Deng, L., Shangguan, Z.P., Sandra, S., 2013. Changes in Soil Carbon and Nitrogen following Land Abandonment of Farmland on the Loess Plateau. China. Plos One 8, e71923.

Duan, J., Liu, Y.J., Tang, C.J., Shi, Z.H., Yang, J., 2021. Efficacy of orchard terrace measures to minimize water erosion caused by extreme rainfall in the hilly region of China: Long-term continuous in situ observations. J. Environ. Manage. 278, 1–15. https://doi.org/10.1016/j.jenvman.2020.111537.

Dubus, I.G., Brown, C.D., 2002. Sensitivity and first-step uncertainty analyses for the preferential flow model MACRO. J. Environ. Qual. 31, 227–240. https://doi.org/10.2134/jeq2002.2270.

Ellison, W.D., 1945. Some effects of raindrops and surface-flow on soil erosion and infiltration. EOS Trans. Am. Geophys. Union 26 (3), 415–429. https://doi.org/10.1029/TR026i003p00415.

Filoso, S., Bezerra, M.O., Weiss, K.C., Palmer, M.A., 2017. Impacts of forest restoration on water yield: a systematic review. PLoS One 12, 1–26. https://doi.org/10.1371/journal.pone.0183210.

Flury, M., Flühler, H., Jury, W.A., Leuenberger, J., 1994. Susceptibility of soils to preferential flow of water: A field study. Water Resour. Res. 30 (7), 1945–1954. https://doi.org/10.1029/94WR00871.

Gargiulo, L., Mele, G., Terribile, F., 2015. The role of rock fragments in crack and soil structure development: a laboratory experiment with a Vertisol. Eur. J. Soil Sci. 66 (4), 757–766. https://doi.org/10.1111/ejss.12263.

Ghestem, M., Sidle, R.C., Stokes, A., 2011. The Influence of Plant Root Systems on Subsurface Flow: Implications for Slope Stability. Bioscience 61 (11), 869–879. https://doi.org/10.1525/bio.2011.61.11.6.

Gumbs, F., Warkentin, B., 1972. The effect of bulk density and initial water content on infiltration in clay soil samples. Soil Sci. Soc. Am. J. 36, 720–724. https://doi.org/10.2136/sssaj1972.03615995003600050014x.

Guo, L., Liu, Y., Wu, G.L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.Q., Tian, F.P., He, H.H., 2019. Preferential water flow: Influence of alfalfa (Medicago sativa L.) decayed root channels on soil water infiltration. J. Hydrol. 578, 124019 https://doi.org/10.1016/j.jhydrol.2019.124019.

Guo, T., Wang, Q., Li, D., Zhuang, J., 2010. Effect of surface stone cover on sediment and solute transport on the slope of fallow land in the semi-arid loess region of northwestern China. Journal of Soils and Sediments 10 (6), 1200–1208. https://doi.org/10.1007/s11368-010-0257-8.

Gutiérrez, F., Gutiérrez (Eds.), M.,, 2016. Landforms of the Earth, (1. ed.:),. Springer International Publishing, pp. 221–236.

Gwenzi, W., Hinz, C., Holmes, K., Phillips, I.R., Mullins, I.J., 2011. Field-scale spatial variability of saturated hydraulic conductivity on a recently constructed artificial ecosystem. Geoderma 166 (1), 43–56. https://doi.org/10.1016/j.geoderma.2011.06.010.

Hua, F.Y., Bruijnzeel, L.A., Meli, P., Martin, P.A., Zhang, J., Nakagawa, S., Miao, X.R., Wang, W.Y., McEvoy, C., Pena-Arancibia, J.L., Brancalion, P.H.S., Smith, P., Edwards, D.P., Balmford, A., 2022. The biodiversity and ecosystem service contributions and trade-offs of forest restoration approaches. Science 376, 839–844. https://doi.org/10.1126/science.abl4649.

Huang, Z., Tian, F.P., Wu, G.L., Liu, Y., Dang, Z.Q., 2017. Legume grasslands promote precipitation infiltration better than gramineous grasslands in arid regions. Land Degrad. Dev. 28, 309–316. https://doi.org/10.1002/ldr.2635.

Jiang, X.J., Liu, W.J., Chen, C.F., Liu, J.Q., Yuan, Z.Q., Jin, B.C., Yu, X.Y., 2018. Effects of three morphometric features of roots on soil water flow behavior in three sites in China. Geoderma 320, 161–171. https://doi.org/10.1016/j.geoderma.2018.01.035.

Jørgensen, P.R., Hoffmann, M., Kistrup, J.P., Bryde, C., Bossi, R., Villholth, K.G., 2002. Preferential flow and pesticide transport in a clay-rich till: field, laboratory, and modeling analysis. Water Resour. Res. 38 (11), 28-1-28-15. https://doi.org/10.1029/2001WR000494.

Kan, X.Q., Cheng, J.H., Hou, F., 2020. Response of Preferential Soil Flow to Different Infiltration Rates and Vegetation Types in the Karst Region of Southwest China. Water. 12, 1778. https://doi.org/10.3390/w12061778.

Kato, H., Onda, Y., Tanaka, Y., Asano, M., 2009. Field measurement of infiltration rate using an oscillating nozzle rainfall simulator in the cold, semiarid grassland of Mongolia. Catena 76 (3), 173–181. https://doi.org/10.1016/j.catena.2008.11.003.

Kim, Y., Darnault, C.J.G., Bailey, N.O., Parlange, J., Steenhuis, T.S., 2005. Equation for describing solute transport in field soils with preferential flow paths. Soil Sci. Soc. Am. J. 69, 291–300. https://doi.org/10.2136/sssaj2005.0291a.

Kottek, M., Grieser, J., Beck, C., Rudolf, B., Rubel, F., 2006. World map of the Köppen-Geiger climate classification updated. Meteorol. Zeitschrift 15, 259–263. https://doi.org/10.1127/0941-2948/2006/0130.

Lipiec, J., Kuś, J., Słowińska-Jurkiewicz, A., Nosalewicz, A., 2006. Soil porosity and water infiltration as influenced by tillage methods. Soil Tillage Res. 89, 210–220. https://doi.org/10.1016/j.still.2005.07.012.

Liu, Y., Cui, Z., Huang, Z., López-Vicente, M., Wu, G.L., 2019. Influence of soil moisture and plant roots on the soil infiltration capacity at different stages in arid grasslands of China. Catena 182, 104147. https://doi.org/10.1016/j.catena.2019.104147.

Liu, J.G., Zhang, H., Fu, J., He, H., Fu, Q.K., Zhu, Y.P., Lan, Y.B., Wang, Y., Ma, M.J., 2014. Periglacial landforms in the Mt. Laotudingzi of eastern Liaoning province: Characteristics and environmental significance. J. Glaciol. Geocryol. 36 (6), 1420–1429. https://doi.org/10.7522/j.issn.1000-0240.2014.0169.

Liu, W.J., Zhu, C.J., Wu, J.E., Chen, C.F., 2016. Are rubber-based agroforestry systems effective in controlling rain splash erosion? Catena 147, 16–24. https://doi.org/10.1016/j.catena.2016.06.034.

Lozano-Baez, S.E., Cooper, M., Meli, P., Ferraz, S., Rodrigues, R.R., Sauer, T.J., 2019. Land restoration by tree planting in the tropics and subtropics improves soil infiltration, but some critical gaps still hinder conclusive results. For. Ecol. Manage. 444, 89–95. https://doi.org/10.1016/j.foreco.2019.04.046.

Luo, Z.T., Niu, J.Z., Xie, B.Y., Zhang, L., Chen, X.W., Berndtsson, R., Du, J., Ao, J.K., Yang, L., Zhu, S.Y., 2019. Influence of root distribution on preferential flow in deciduous and coniferous forest soils. Forests 10 (11), 986. https://doi.org/10.3390/f10110986.

Luo, Z.T., Niu, J.Z., He, S.Q., Zhang, L., Chen, X.W., Tan, B., Wang, D., Berndtsson, R., 2022. Linking roots, preferential flow, and soil moisture redistribution in deciduous and coniferous forest soils. J. Soil. Sediment. 23 (3), 1524–1538. https://doi:10.1007/s11368-022-03375-w.

Ma, D.H., Shao, M.A., 2008. Determination error of water content in stony soil. Acta Pedol. Sin. 2, 201–206. https://doi.org/10.3321/j.issn:0564-3929.2008.02.002.

Ma, M.Y., Wang, Q.M., Liu, R., Zhao, Y., Zhang, D.Q., 2023. Effects of climate change and human activities on vegetation coverage change in northern China considering extreme climate and time-lag and -accumulation effects. Sci. Total Environ. 860, 160527 https://doi.org/10.1016/j.scitotenv.2022.160527.

Mei, X., Zhu, Q., Ma, L., Zhang, D., Wang, Y., Hao, W., 2018. Effect of stand origin and slope position on infiltration pattern and preferential flow on a Loess hillslope. Land Degrad. Dev. 29 (5), 1353–1365. https://doi.org/10.1002/ldr.2928.

Merdun, H., Meral, R., Demirkıran, A.R., 2008. Effect of the initial soil moisture content on the spatial distribution of the water retention. Eurasian Soil Sci. 41 (10), 1098–1106. https://doi.org/10.1134/S1064229308100128.

Meyer, L.D., Johnson, C.B., Foster, G.R., 1972. Stone and woodchip mulches for erosion control on construction sites. J. Soil Water Conserv. 27 (6), 264–269. https://doi.org/10.1016/0016-7061(72)90010-9.

Morales, V.L., Parlange, J.Y., Steenhuis, T.S., 2010. Are preferential flow paths perpetuated by microbial activity in the soil matrix? A review. J. Hydrol. 393, 29–36. https://doi.org/10.1016/j.jhydrol.2009.12.048.

Murphy, C.P., Banfield, C.F., 1978. Pore space variabaity in a sub-surface horizon of two soils. Eur. J. Soil Sci. 29 (2), 156–166. https://doi.org/10.1111/j.1365-2389.1978.tb02046.x.

Nair, P.R., Nair, V.D., Kumar, B.M., Showalter, J.M., 2010. Carbon sequestration in agroforestry systems. Adv. Agron. 108, 237–307. https://doi.org/10.1016/S0065-2113(10)08005-3.

Nasri, B., Fouché, O., Torri, D., 2015. Coupling published pedotransfer functions for the estimation of bulk density and saturated hydraulic conductivity in stony soils. Catena 131, 99–108. https://doi.org/10.1016/j.catena.2015.03.018.

Novák, V., Kňava, K., Šimůnek, J., 2011. Determining the influence of stones on hydraulic conductivity of saturated soils using numerical method. Geoderma 161 (3–4), 177–181. https://doi.org/10.1016/j.geoderma.2010.12.016.

Nyssen, J., Haile, M., Poesen, J., Deckers, J., Moeyersons, J., 2006. Removal of rock fragments and its effect on soil loss and crop yield, Tigray. EthiopiaJ. Soil Use Manag. 17 (3), 179–187. https://doi.org/10.1111/j.1475-2743.2001.tb00025.x.

Pierret, A., Latchackak, K., Chathanvongsa, P., Sengtaheuanghoung, O., Valentin, C., 2007. Interactions between root growth, slope and soil detachment depending on land use: a case study in a small mountain catchment of Northern Laos. Plant and Soil 301 (1–32), 51–64. https://doi.org/10.1007/s11104-007-9413-3.

Poelking, E.L., Schaefer, C.E.R., Fernandes, E.I., de Andrade, A.M., Spielmann, A.A., 2015. Soil-landform-plant-community relationships of a periglacial landscape on Potter Peninsula, maritime Antarctica. Solid Earth 6 (2), 583–594. https://doi.org/10.5194/sed-6-2261-2014.

Poesen, J., 1986. Surface sealing as influenced by slope angle and position of simulated stones in the top layer of loose sediments. Earth Surf Process Landf. 1986, 11 (1): 1-10. https://doi.org/10.1002/esp.3290110103.

Porporato, A., D'odorico, P., Laio, F., Ridolfi, L., Rodriguez-Iturbe, I., 2002. Ecohydrology of water-controlled ecosystems. Adv. Water Resour. 25 (8–12), 335–1348. https://doi.org/10.1016/S0309-1708(02)00058-1.

Qi, W., Zhang, Z.Y., Wang, C., Chen, Y., Zhang, Z.M., 2020. Crack closure and flow regimes in cracked clay loam subjected to different irrigation methods. Geoderma 358, 113978. https://doi.org/10.1016/j.geoderma.2019.113978.

Qiu, D.X., Xu, R.R., Wu, C.X., Mu, X.M., Zhao, G.J., Gao, P., 2023. Effects of vegetation restoration on soil infiltrability and preferential flow in hilly gully areas of the Loess Plateau. China. Catena. 221 (A), 106770 https://doi.org/10.1016/j.catena.2022.106770.

Ritsema, C.J., Dekker, L.W., 2000. Preferential flow in water repellent sandy soils: principles and modeling implications. J. Hydrol. s 231–232 (1), 308–319. https://doi.org/10.1016/S0022-1694(00)00203-1.

Rodríguez, F., Mayor, A.G., Rietkerk, M., Bautista, S., 2018. A null model for assessing the cover-independent role of bare soil connectivity as indicator of dryland functioning and dynamics. Ecol. Ind. 94, 512–519. https://doi.org/10.1016/j.ecolind.2017.10.023.

Schwärzel, K., Ebermann, S., Schalling, N., 2012. Evidence of double-funneling effect of beech trees by visualization of flow pathways using dye tracer. J. Hydrol. 470–471, 184–192. https://doi.org/10.1016/j.jhydrol.2012.08.048.

Sheng, F., Liu, H., Wang, K., Zhang, R., Tang, Z., 2014. Investigation into preferential flow in natural unsaturated soils with field multiple-tracer infiltration experiments and the active region model. J. Hydrol. 508, 137–146. https://doi.org/10.1016/j.jhydrol.2013.10.048.

Stokes, A., Atger, C., Bengough, A.G., Fourcaud, T., Sidle, R., 2009. Desirable Plant root traits for protecting natural and engineered slopes against landslides. Plant and Soil 324 (1), 1–30. https://doi.org/10.1007/s11104-009-0159-y.

Sun, D., Yang, H., Guan, D.X., Yang, M., Wu, J.B., Yuan, F.H., Jin, C.J., Wang, A.Z., Zhang, Y.S., 2018. The effects of land use change on soil infiltration capacity in China: A meta-analysis. The Science of the Total Environment 626, 1394–1401. https://doi.org/10.1016/j.scitotenv.2018.01.104.

Valentin, C., Casenave, A., 1992. Infiltration into Sealed Soils as Influenced by Gravel Cover. Soil Sci. Soc. Am. J. 56 (6), 1667–1673. https://doi.org/10.2136/sssaj1992.03615995005600060002x.

Van Noordwijk, M., Brouwer, G., Meijboom, F., do Rosario, M., Oliveira, G., Bengough, A.G., 2000. Trench Profile Techniques and Core Break Methods. In Root Methods: A Handbook; Smit, A.L., Bengough, A.G., Engels, C., Noordwijk, M., Pellerin, S., Geijn, S.C., Eds.; Springer: Berlin, Germany. pp. 211-223. https://doi.org/10.1007/978-3-662-04188-8_7.

Wang, D., Niu, J.Z., Miao, Y.B., Yang, T., Berndtsson, R., 2023. Predicting Soil Saturated Water Conductivity Using Pedo-Transfer Functions for Rocky Mountain Forests in Northern China. Forests 14, 1097. https://doi.org/10.3390/f14061097.

WRB, 2015. World Reference Base for Soil Resources 2014. Update 2015. World Soil Resources Reports No. 106. Food and Agriculture Organization of the United Nations, Rome, 192 p.

Zhang, H., Liu, J.G., Fu, J., Sun, M.Y., Luo, X., He, H., Zhu, Y.P., Lan, Y.B., Wang, Y., Ma, M.J., 2015. Plant community types and basic characteristic in the periglacial areas of the Mt.Laotudingzi, eastern Liaoning mountainous regions. J. Glaciol. Geocryol. 37 (2), 500–510. https://doi.org/10.7522/j.issn.1000-0240.2015.0057.

Zhang, Y.H., Zhang, M.X., Niu, J.Z., Zheng, H.J., 2016a. The preferential flow of soil: A widespread phenomenon in pedological perspectives. Eurasian Soil Sci. 49 (6), 661–672. https://doi.org/10.1134/S1064229316060120.

Zhang, Y., Zhang, M., Niu, J., Li, H., Xiao, R., Zheng, H., Bech, J., 2016b. Rock fragments and soil hydrological processes: significance and progress. Catena 147, 153166. https://doi.org/10.1016/j.catena.2016.07.012.

Zhang, W.J., Zhu, X.A., Ji, S.Y., Chen, C.F., Zeng, H.H., Zou, X., Yang, B., Jiang, X.J., Liu, W.J., 2022. Soil water movement differences relating to banana (Musa nana Lour.) plantation regime. Land Degrad. Dev. 33 (11), 1821–1835. https://doi.org/10.1002/ldr.4264.

Zhou, B.B., Shao, M.A., Shao, H., 2009. Effects of rock fragments on water movement and solute transport in a Loess Plateau soil. C. R. Geosci. 341 (6), 462–472. https://doi.org/10.1016/j.crte.2009.03.009.

Zhou, B.B., Shao, M.A., Wang, Q.J., Yang, T., 2011. Effects of Different Rock Fragment Contents and Sizes on Solute Transport in Soil Columns. Vadose Zone J. 10 (1), 386–393. https://doi.org/10.2136/vzj2009.0195.

Zhu, X.A., Chen, C.F., Wu, J.N., Yang, J.B., Zhang, W.J., Zou, X., Liu, W.J., Jiang, X.J., 2019. Can intercrops improve soil water infiltrability and preferential flow in rubber-based agroforestry system? Soil Tillage Res. 191, 327–339. https://doi.org/10.1016/j.still.2019.04.017.